Logophors, Person and the Typological Landscape of Shiftiness*

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Abstract

Logophoric pronouns of the kind found in West African languages have been extensively studied by typologist and formal linguists alike for more than five decades. However, there has been few attempts to connect this line of work with a formal approach to pronouns and the features they consist of, most notably Person (Silverstein, 1976; Zwicky, 1977; Noyer, 1992; Harbour, 2016). The overarching goal of this contribution is to fill this gap, and provide a formal account of logophoric pronouns grounded in a full-fledged theory of person features. Taking inspiration from Schlenker (2003), we propose that logophoric pronouns are non-indexical first persons, specified with a -ACTUAL feature. Our account is able to derive a wide range of distributional and interpretive constraints of logophors across languages, most notably i) their *de se* readings, ii) their agreement patterns, and iii) their distribution in embedded environments, while preserving a conservative, property-based account of features, which we argue are best conceived as binary (Harbour, 2011a, 2013). In addition, we provide arguments for a conception of alternatives in which structural complexity (in the sense of Katzir 2007) is assessed in the morphosyntax, at the level of features.

1 Introduction

Logophoric pronouns (henceforth, LPs) are pronominal forms found in the paradigms of a wide range of sub-saharian languages, mostly stemming from the Chadic, Niger-Congo and Nilotic families, the function of which is to refer to the producer of a speech or thought, (1):[1]

*Ackowledgements to be added.

[1] Here is a list of the glossing conventions used in this paper: AOR = aorist, LOG = logophoric pronoun, PRFVE = perfective aspect, QUOT = quotative particle, REP = reportative, 1 = first person, 2 = second person, 3 = third person, ABS = absolutive, ACC = accusative, AUX = auxiliary, COMP = complementizer, COP = copula, DAT = dative, DEF = definite, DEM = demonstrative, ERG = ergative, F = feminine, FOC = focus, FUT = future, INS = instrumental, IRR = irrealis, M = masculine, NEG = negative, NOM = nominative, OBJ = object, OBL = oblique, PFV = perfective, PL = plural, POSS = possessive, PRF = perfect, PROG = progressive, PRS = present, PST = past, SBJV = subjunctive, SG = singular.

(1) Kofi be **yè** dzo
Kofi say LOG leave
'Kofi$_i$ said that he$_{i/*j}$ left.'

[Ewe (Niger-Congo), Clements 1975]

Observing, based on pioneering comparative studies by e.g. Clements (1975); Culy (1994a, 1997); Von Roncador (1992) and Curnow (2002b), that the distribution of LPs is mostly restricted to the complement of attitude predicates, previous analyses in the generative tradition modelled them as obligatory-bound variables specified with a formal LOG feature (Koopman and Sportiche, 1989; Heim, 2001, 2002; von Stechow, 2002, 2003; Pearson, 2015). However, there has been almost no attempts (a notable exception being Deal (2024), discussed in detail in §3) to consider these theoretical assumptions in the light of a broader stream of research on the person features of pronominal systems (Silverstein, 1976; Zwicky, 1977; Noyer, 1992; Harley and Ritter, 2002; Harbour, 2003; McGinnis, 2005; Nevins, 2007; Sauerland, 2008; Harbour, 2016; Sauerland and Bobaljik, 2022). An immediate question pressed on anyone trying to pursue such a goal is the following: assuming that all known person systems can be derived with a combination of only two person primitives, $\pm$AUTH and $\pm$PART (or closely-defined analogues, cf. §4.5.2), what are the person specifications of LPs? If not a person, what is the status of LOG as a feature? Put in a different, less theory-laden way: what constitutes an adequate representation of LPs within person paradigms of logophoric languages? This article aims at providing a rather conservative answer to that question: LPs are, inherently, non-indexical first- (or second-, for addressee-oriented LPs) person forms, lexicalizing feature bundles similar to that of their indexical counterparts, except for a negatively-specified feature -ACTUAL (inspired from Schlenker 2003), which restricts their interpretation to non-actual contexts – contexts in which the reported and actual speakers are distinct, capturing most of their distribution in the languages at stake.

The rest of this article spells out this proposal and is structured as follows. §2 surveys the empirical landscape of LPs and their main distributional and interpretive properties, that any accurate theory of such pronouns need to account for. §3 presents three influential accounts of LPs: the 'standard' binding account (Heim, 2001, 2002; von Stechow, 2002, 2003; Anand, 2006; Pearson, 2015; Deal, 2018), according to which LPs are obligatory bound variables subject to syntactic locality restrictions; the featural account of Deal (2024), in which LPs are elements specified with a dedicated person feature, AUTH-I, referring to attitude centers; and the presuppositional account of Bassi et al. (2025), which improves on the standard account by assuming that LPs are specified with a presuppositional LOG feature that can be ignored in contexts involving alternatives. We then argue that these analyses all partially fail to account for the empirical data at stake, and in §4, we introduce our alternative account, which retains essential components of its predecessors but improves in empirical coverage, while being more conservative about the ontology of person it assumes. We also argue that the data under consideration, notably the various distributional patterns of LPs, warrant an analysis in which LPs compete with other forms in the paradigm via a complexity metric (Katzir, 2007; Fox and Katzir, 2011) that evaluates structures at the morphosyntactic level, not the lexicon, with important consequences for our theoretical conception of features. §5 concludes.

2 Logophoric pronouns: main features

This section introduces logophoric pronouns and their properties, including their distributional restriction to embedded clauses headed by attitude verbs (§2.1), their preference for *de se* interpretation (§2.2), their ability to license agreement mismatches (§2.3) as well as disjointness inferences (§2.4).

2.1 Distribution and licensing

In logophoric languages, a dedicated pronominal form is used in attitude reports to cross-reference the author of the report:[2]

(2) a. Oumar Anta **inyemɛn** waa be gi
Oumar Anta **LOG.ACC** seen AUX said
'Oumar$_i$ said that Anta had seen him$_i$'

b. Oumar Anta **won** waa be gi
Oumar Anta **3SG.ACC** seen AUX said
'Oumar$_i$ said that Anta had seen him$_{*i/k}$'
[Donno Sɔ (Niger-Congo, Dogon; Mali), Culy 1994a: (1)]

As (2b) and (3) illustrate, third-person pronouns cannot co-refer with the attitude holder introduced in the matrix sentence, and therefore cannot substitute for LPs:

(3) Kofi be **e** dzo
Kofi say **3SG** leave
'Kofi$_i$ said that he$_{\{\#i/j\}}$ left'
[Ewe, Clements 1975]

In some cases, the logophoric form can be realized as a verbal affix, as in the language Akoose (Niger-Congo, Bantu; Cameroon), (4):

(4) a. à hɔbé ǎ mə-kàg
3SG said REP LOG-should.go
'He$_i$ said he$_{i/*j}$ should go.'

b. à hɔbé ǎ á-kàg
3SG said REP 3SG-should.go
'He$_i$ said he$_{*i/j}$ should go.'
[Akoose, Hedinger 1984: 95]

[2] In this paper, we use the terms 'logophoric pronoun', 'logophoric form' and 'logophor' interchangeably to refer to the elements traditionally designated as such in the seminal sense of Hagège (1974), and not to other forms that have been labeled 'logophors' with different theoretical implications in the generative literature, such as long-distance anaphors (i. e., anaphors that seem not to obey the 'condition A' principle of Binding Theory; see Kuno (1987), Reinhart and Reuland (1993), Charnavel and Sportiche (2016) i.a.). For arguments that the former class is not reducible to the latter, see among others Culy (1994a); Dimmendaal (2001) and Bassi et al. (2025).

The distribution of logophoric forms is somewhat restricted, being confined to a certain class of syntactic-semantic environment, namely, finite clauses involving attitude predicates.[3] As first noted by Culy (1994a), out of a sample of 48 logophoric languages, 29 would allow LPs to appear under *say*, while only a subset of this group (13) would allow LPs to appear under *think*; the same goes for *know*, where LPs are licensed for another subset of 6 languages out of the sample. This allows Culy (1994a) to conclude that LPs are licensed by a hierarchically-ordered set of attitude predicates forming an implicational scale: if a given language licenses LPs under any element in the scale, then it must also license them under any element to its right.[4]

(5) **A hierarchy of logophoric licensers** [Culy 1994a: (10)]
speech < thought < knowledge < direct perception

Although logophoric pronouns mainly occur in syntactic contexts involving one finite embedded clause in which the LP cross-references another NP introduced in the matrix clause, they are able to co-refer to elements in more than one clause up in multiple-embedding configurations:

(6) a. Kofi xɔ-e se be Ama gblɔ be yè-ʃu yè.
Kofi receive hear COMP Ama say COMP LOG-beat LOG
'Kofi$_i$ believed that Ama$_j$ said that LOG$_{i/j}$ beat LOG$_{i/j}$.'
[Ewe, Clements 1975, 73]

b. Marie be Kofi xɔse be yè na yè cadeau.
Mary say Kofi believe COMP LOG give LOG gift
'Mary$_i$ said that kofi$_j$ believed that LOG$_{i/j}$ gave LOG$_{i/j}$ a gift.'
[Ewe, Pearson 2015, (47)]

(7) a. Olù ní Adé so pé àwon rì òun
Olu say Ade say COMP 3SG.PL see LOG
'Olu$_i$ said that Ade$_j$ said that they$_{j,k}$ saw him$_k$.'
[Yoruba (Niger-Congo: Yoruboid), Adesola 2005, 207]

b. Olù so pé Ade ro pé bàbá oun ti rì ìyá òun
Olu say COMP Ade think COMP father LOG PERF see mother LOG
'Olu$_i$ said that Ade$_j$ said that his$_{i/j}$ father saw his$_{i/j}$ mother.'
[Yoruba, Anand 2006, 60]

This latter set of data is especially hard to accommodate if one considers that the presence of an LP is subject to syntactic locality restrictions; see §3.

2.2 *De se* interpretation

Attitudes *de se* are a distinct subtype of attitudes that involve first-personal or 'self-locating' beliefs (Perry, 1977; Lewis, 1979; Chierchia, 1989; Anand, 2006; Pearson,

[3]Although this requirement does not seem absolute, with certain languages allowing LPs to appear outside of such environments, e.g. intention or purpose clauses (Nikitina, 2020, 2023).

[4]It is unclear why Culy (1994a) includes the class of 'direct perception' verbs within his hierarchy, since he explicitly mentions that no language seems to license LPs under this category. We are reproducing the original proposal, without modifications.

2012). Typically, the content of a *de se* attitude can be felicitously attributed to an agent if they relate to that content in a first-personal way, i.e., recognize that they are the experiencer of that content. As first observed by Clements (1975) for Ewe, LPs favor a *de se* interpretation, rendering them infelicitous in non-*de se* scenarios in which the attitude holder is unaware that the content of the report is about himself, as (8) and (9) illustrate for Ewe and Ibibio, respectively:

(8) *Context: an Asian woman was declared missing from a party touring the Eldgjá volcanic region in south Iceland after getting off the party's bus to freshen up. She only hopped off the bus briefly, but had also changed her clothes - and her fellow travelers did not recognize her when she climbed back on again to continue the party's journey. When the details of the missing person were issued, the woman reportedly didn't recognize her own description [woman with a pink sweater] and unwittingly joined the search party for herself.*

a. Asia nyɔnu la xɔese be é bú
   Asian woman DEF believe.3SG COMP 3SG be
   'The asian woman$_i$ believes that she$_i$ is lost.'

b. #Asia nyɔnu la xɔese be yè bú
   Asian woman DEF believe.3SG COMP LOG be
   'The asian woman$_i$ believes that she$_i$ is lost.'

[Ewe, Bimpeh 2019: (15-16)]

(9) *Context: Ekpe sings on occasion, but will never admit that he is any good. So one time, during one of his performances, you record him without his knowledge. Some time later, you play back the recording to him without telling him who is singing. Ekpe doesn't recognize himself in the recording, and comments "he sings well."*

a. Ekpe a-bo ke anye a-diyono ikwo ikwo mfonmfon
   Ekpe 3SG-say COMP 3SG 3SG-know sing song well
   'Ekpe$_i$ said that he$_{i,j}$ sings well.'

b. #Ekpe a-bo ke imo i-me i-diyono ikwo ikwo mfonmfon
   Ekpe 3SG-say COMP LOG LOG-PRS LOG-know sing song well
   'Ekpe$_i$ said that he$_i$ sings well.'

[Ibibio (Niger-Congo; Southern Nigeria), Newkirk 2019: (11)]

In both scenarios, the author of the report fails to recognize herself as such - i.e., fails to self-ascribe the property of being the author of the report; as a consequence, a LP cannot be used, and a 3rd person form is licensed for *de re* cross-reference.

While it has been argued that LPs are unambiguously read *de se* in various languages such as Ewe, Ibibio, Tangale and Yoruba (Schlenker, 1999, 2003; Haida, 2009; Newkirk, 2019; Bimpeh, 2019; Bimpeh and Sode, 2021; Bassi et al., 2025), it has been observed that they allow for a *de re* intepretation in varieties of Ewe and Yoruba (Adesola, 2006; Pearson, 2015), providing evidence for cross-linguistic variation in this respect. In what follows, we will simply assume (following Pearson 2015) that the default meaning of LPs is *de se*, with potential *de re* readings being derived with the help of additional semantic machinery: see §4 for further discussion.

2.3 Agreement mismatches

In some languages, LPs trigger first person agreement on the embedded verb. This is illustrated in (2) for Donno Sɔ:

(10) a.

Oumar	inyemɛ	jɛmbɔ	paza	**bolum**	miñ	tagi
Oumar	LOG	sack.DEF	drop	left.1SG	1SG.OBJ	inform.PST

‘Oumar$_i$ told me that he$_i$ had left without the sack.’

b.

Oumar	ma	jɛmbɔ	paza	**boli**	miñ	tagi
Oumar	1SG.SBJV	sack.DEF	drop	left.3SG	1SG.OBJ	inform.PST

‘Oumar$_i$ told me that I had left without the sack.’

[Donno Sɔ, Culy 1994b: (20)]

In (10a), the embedded verb is inflected for the first person, in spite of the agreement controller being the logophoric form *inyemɛ*, and not a first person pronoun. Moreover, note that first person agreement for the embedded verb is optional here, and that ‘standard’, third person agreement as in (10b) results in a disjoint reading, in which the agent of leaving and Oumar have to be distinct individuals.

In yet some other subset of languages, one can find a dedicated logophoric agreement morphology surfacing on the verb with singular antecedents only; with plural antecedents, agreement morphology has to be first person, just like in Donno Sɔ. This is the case of languages Ibibio, Ekpeye and Efik (all three Cross-River, Nigeria and Cameroon), in which the plural logophoric agreement marker is syncretic with the plural first person form.

(11) a.

Ekpe	a-bo	ke	imo	ì-ma	í-to	Udo
Eke	3SG.say	COMP	LOG	LOG-PST	LOG-hit	Udo

‘Ekpe$_i$ says that he$_i$ hit Udo.’

b.

ommo	e-ke	e-bo	ke	mmimo	ì-ma	ì-kot	nwet
3PL	3PL-PST	3PL-say	COMP	LOG.PL	1PL-PST	1PL-read	book

‘They$_i$ said that they$_i$ read a book.’

[Ibibio, adapted from Newkirk 2019: (1a)-(5)]

(12) a.

ù-kà	ɓú	yá’	zè
3SG-say.PST	COMP.NON1	LOG.SG	go.PST

‘He$_i$ said that he$_i$ went.’

b.

ù-kà-ɓɛ	ɓú	à-zè
3SG-say.PST-PL	COMP.NON1	1PL-go.PST

‘They$_i$ said that they$_i$ went.’

[Ekpeye, adapted Curnow 2002b: (27)-(29), after Clark 1972]

Another category of logophoric elements is what Curnow (2002b) dubs ‘first-person logophoricity’ for a large class of Nilo-Saharan languages (see also Culy 1994b). In these languages, no dedicated logophoric form is present, but logophoricity is realized via first person agreement on the embedded verb. This is exemplified in (13) for the language Karimojong (Nilotic), where a third person subject triggers first person agreement in the embedded clause:

(13) àbu papà tolim ɛbè àlózì iŋèz morotó
AUX father say COMP 1SG.go.NPST 3SG Moroto
'Father$_i$ said that he$_i$ was going to Moroto.'
[Karimojong, Curnow 2002b: (18)]

Under standard assumptions about the semantics of first person, the data above is quite surprising: it seems that logophoricity and indexicality are conflated in languages such as Donno Sɔ, Ibibio and Ekpeye, in which first-personal markers appear in spite of being anteceded by non-actual speakers. This tight relationship between the two categories has, in our opinion, been underexplored – Deal (2024) being a notable exception, that we discuss in detail in §3.2.

2.4 Disjointness effects

In logophoric languages, the use of a non-logophoric form prevents co-reference with the reported speaker: in Aghem (Niger-Congo; Cameroon), for instance, the use of the 3rd person pronoun *ù* instead of the logophoric form *é* indicates that its referent is not the reported speaker, Nsen, but some other, salient female individual:

(14) a. Nnsɨni dzɛ enyɨa é bvʉ nù
Nsen say COMP LOG fall FOC
'Nsen$_i$ said that she$_i$ fell.'

b. Nnsɨni dzɛ enyɨa ù bvʉ nù
Nsen say COMP 3SG fall FOC
'Nsen$_i$ said that she$_{*i/j}$ fell.'
[Aghem, Butler 2009: (10-11)]

Examples (2b) and (3) make the same point for Donno Sɔ and Ewe, respectively. Following Marty (2018), we will refer to this pattern as a *disjointness inference*:

(15) **Disjointness inference** [Marty 2018: fn.1]
A nominal expression α is interpreted as disjoint from a nominal expression β if the interpretation of α does not — exhaustively or partially — co-refer with or co-vary with that of β.

For languages with an LP in their paradigm, the generalization seems to be that using a 3rd person form *in lieu* of an LP will trigger the inference that the referent of the 3rd person is not the attitude holder defined by the logophoric domain. However, this fails to capture the infelicity of LPs to cross-refer the author of the report in *de re* scenarios such as (8) and (9) above. Rather, the generalization seems to be sensitive not to co-reference between individuals, but rather between *world-individual pairs*; a LP is felicitous in a logophoric context only if it matches to the correct individual (the attitude holder) in all the worlds quantified over by the logophoric verb. The generalization is therefore intensional:

(16) **Disjointness inference in logophoric contexts**
In a language L where an LP is licensed under an approppriate attitude verb A_c, an embedded 3rd person proform cannot co-refer with the attitude holder if it is non-*de se*, i.e. if there is at least one world in which the center of the attitude fails to recognize herself as such.

Similarly, as initially noted by Hyman and Comrie (1981) for Gokana, LPs generally cannot take 1st person pronouns as antecedents. In other words, for a given speech report, when the reported and current speaker are intended to be co-referential, a logophor cannot be used.[5] We refer to this as the *1-LOG pattern:

(17) a. aè kɔ aè dɔ-ɛ
3SG said 3SG fell-LOG
'He$_i$ said he$_i$ fell'

b. mm kɔ mm dɔ
1SG said 1SG fell
'I$_i$ said I$_i$ fell'

c. #mm kɔ mm dɔ-ɛ
1SG said 1SG fell-LOG
'I$_i$ said I$_i$ fell'
[Gokana (Niger-Congo; Ogoni, Nigeria) Hyman and Comrie 1981: (2)-(11)]

(18) a. Kofí ŋa bə yi lɔ Áma
Kofi know COMP LOG love Ama
'Kofi$_i$ knows that he$_{i/*j}$ loves Ama'

b. #ɱə ŋa bə yi lɔ Áma
1SG know COMP LOG love Ama
Intended: 'I$_i$ know that I$_i$ love Ama'
[Danyi Ewe (Niger-Congo, Togo); O'Neill 2015: (3a, c)]

A similar pattern can be found in Wan (Niger-Congo, Ivory Coast; Nikitina 2012a), various varieties of Ewe (Pearson 2015, Bimpeh 2019), as well as Ibibio (Newkirk, 2019).

Last, a variety of LP languages exhibit a special case of 'person neutralization' between third and second person; as a consequence, logophors can take second person antecedents as well as third, with singular and plural number features alike. As already discussed, first person antecedence is excluded:

(19) a. là gé ɓà súglù é lɔ
2SG said LOG.SG Manioc DEF ate
'You$_i$ said you$_i$ had eaten the manioc.'

b. à gé mɔ kú má
2PL said LOG.PL house EQUAT
'You$_i$ said it was your$_i$ house.' [Wan, Nikitina 2012a: (5a, b)]

The phenomenon is actually broader, extending to various other reference-tracking systems, such as those found in the Sino-Tibetan languages Jingpho (Zu, 2018) and Newar (Coppock and Wechsler, 2018), as well as languages from the Himalayas, the Caucasus, the Andes, and Highlands New Guinea (San Roque et al., 2017).

It is reasonable to asssume that the rationale behind all the above cases is that the LP competes with other pronouns in similar distribution in a given language. However, formulating a theory of such competition is non-trivial, since it requires an adequate theory

[5] Hyman and Comrie (1981) make a less stronger claim, stating only that (17b) is preferred over (17c).

of logophoric pronouns and their features, and how such features interact with other features in the person domain. In §4, we propose such a theory, which we argue is capable of deriving the various patterns of disjointness inferences exposed in this section.

3 Previous analyses

In this section, the most prominent analyses of logophoric pronouns are introduced: the binding analysis of Heim (2001, 2002) and von Stechow (2002, 2003) (§3.1), as well as two recent accounts which significantly improve on the binding analysis, Deal (2024) (§3.2) and Bassi et al. (2025) (§3.3). We discuss these analyses in turn and show that, although they each put forward crucial elements necessary to capture the semantics of LPs, both fail to capture the full range of data presented in §2, ultimately calling for another analysis (§4).

3.1 The Heim-von Stechow analysis

Under the Heim-von Stechow (HsV) analysis, logophoric pronouns are obligatorily-bound elements consisting of an individual variable (Heim 2001, 2002; von Stechow 2002, 2003; Pearson 2015 i.a.). Most of these analyses are framed within an intensional system whereby attitude verbs are viewed as quantifiers over centered worlds, i. e. world-individual pairs conceived as tuples of type $\langle s, e\rangle$ (Lewis, 1979; Chierchia, 1989). Within such a system, LPs are considered a special kind of pronoun that unambiguously pick up the center of the world it is evaluated against, the individual that the referent takes himself to be in the world of evaluation. Under this analysis, attitude verbs introduce λ-abstractors for individuals in the left of their complements, and LPs come endowed with a syntactic feature LOG that forces them to be bound by this abstractor. To illustrate, the Ewe sentence in (20a) willl be interpreted as in (20d):

(20) a. Kofi be **yè** dzo
Kofi say **LOG** leave
'Kofi$_i$ said that he$_{i/*j}$ left.'

b. $[\lambda w_1.[w_1$ Kofi said$_{[\text{LOG}]}$ that $[\lambda x^2_{[\text{LOG}]}.\lambda w_3.[w_3$ LOG$^2{}_{[\text{LOG}]}$ left $]]]]$

c. $[\![$ LOG left $]\!]^{g,c} = \lambda w.\lambda x.x$ left in w

d. $[\![$ Kofi said that LOG left $]\!]^{g,c} = \lambda w.\forall < w', y > \in \text{SAY}_{K,w}, y$ left in w'

This analysis ensures the *de se* interpretation of LPs (§2.2): since the pronoun is obligatory bound by the individual abstractor, it will unambiguously denote the center of the world-individual pair, that is, the individual that Kofi identifies himself with in his SAY-worlds counterparts, ruling out both non-coreferential and *de re* readings of LOG. It has, however, two important shortcomings. The first one, discussed in detail by Bassi et al. (2025), is empirical and concerns the unability of the theory to derive 'sloppy' (i.e., unbound) readings of LPs in alternative-sensitive environments such as ellipsis or under the scope of focus-associated *only*, (21)-(22):

(21) Élì lè mʉ-kpʉ-m bé yè á dè Àblá. Yàó hã.
Eli COP path-see-PROG COMP LOG IRR marry Abla. Yao too.
'Eli$_i$ hopes that he$_i$ will Marry Abla. Yao$_j$ does ⟨ ... ⟩ too.'
✓ Yao hopes that Yao will marry Abla, too. (sloppy reading)
✓'Yao hopes that Eli will marry Abla, too. (strict reading)

(22) Élì kò yé súsú bé yè dùdzí lè àwù-dódó fé hòwíwlí mè.
Eli only FOC think COMP LOG win in-dress wear POSS contest inside
'Only Eli thinks that he won the costume contest.'
✓ No one$_i$ but Eli think they$_i$ won the costume contest. (sloppy reading)
✓ No one but Eli$_j$ think he$_j$ won the costume contest. (strict reading)
[Bassi et al. 2025: (13)-(9)]

As Bassi et al. (2025) argue, the standard binding account does not predict such readings: the logophor being in each case an obligatory bound variable, there is no way for it to be free in either the elided clause or in the focus alternatives:

(23) **LF of (21):**
Eli$_i$ hopes [$\lambda x^2{}_{[log]}$ that LOG$^2{}_{[log]}$ will Marry Abla]. Yao$_j$ does ⟨ hope [$\lambda x^2{}_{[log]}$ that LOG$^2{}_{[log]}$ will Marry Abla ⟩ too.'

(24) **LF of (22):**
a. Only Eli$_{[F]}$ thinks [$\lambda x^2{}_{[log]}$ that LOG$^2{}_{[log]}$ won the costume contest].
b. ALT(22): $\left\{ \begin{array}{c} \text{Koku}_{[F]} \text{ thinks } [\lambda x^2{}_{[log]} \text{ that LOG}^2{}_{[log]} \text{ won the costume contest}]. \\ \text{Kofi}_{[F]} \text{ thinks } [\lambda x^2{}_{[log]} \text{ that LOG}^2{}_{[log]} \text{ won the costume contest}]. \\ \ldots \end{array} \right\}$

From this, Bassi et al. (2025) conclude that the HvS account of LPs fail to account for such cases and therefore should be amended.

The second shortcoming is theoretical: what is the nature of the syntactic feature LOG? As a technical device, it is necessary in order to ensure a systematic connection between the variable-denoting pronoun and the binder introduced by the attitude verb. However, as a linguistic object, its status is far from being obvious. What kind of feature does LOG refer to? Is it a person feature? If not, what is its kind? This question is addressed in §4.5.2.

3.2 Deal (2024)

Deal (2024) proposes an innovative account in which LPs are pronouns specified with a new kind of first-person feature, AUTH-I, which denotes the center of the index introduced by the attitude predicate. Deal's main claim is not directly about LPs but is motivated by another, closely related phenomenon of 'monstrous agreement', in which agreement markers are realized as first person in clauses embedded by attitude verbs, (25)-(27):

(25) Raju [tanu parigett-ææ-nu ani] cepp-ææ-Du.
Raju 3SG run-PST-1SG COMP say-PST-3SG.M
'Raju$_i$ said that he$_i$ ran.'
[Telugu (Dravidian), Messick 2023: (10b)]

(26) Alsu ***pro*** ber kajčan da miŋga bag-m-a-s-mɤn diep
Alsu ***pro*** one when nPCL 1SG.DAT look.at-NEG-ST-POT-1SG COMP
bel-ä
know.ST-IMPF
'Alsu$_i$ knows that **I**$_i$ would never look at **me**$_{s(c)}$'
[Mishar Tatar (Turkic), Podobryaev 2014: (210)]

In (25) and (26), third person forms can trigger first-person agreement in embedded clauses, just like languages with first-person logophoricity such as Karimojong, (13). Deal (2024) convincingly argues that such cases cannot be assimilated to apparently similar instances of indexical shift, in which first- and second-person indexicals are shifted in embedded clauses in a similar sense (Schlenker 2003; Anand 2006; Deal 2020b; cf. §4.4), since these pronouns do not shift in these languages, as evidenced in (26) and (27):

(27) a. boris man-a ***pro*** san-ba ëç̡l-e-p te-ze kala-rj-ə
boris I.OBJ ***pro*** 2SG-INS work-NPST-1SG say-COMP say-PST-3SG
'Boris$_i$ told me that I / he$_i$ will work with you$_{a(c)}$.'

b. boris man-a ep san-ba ëç̡l-e-p te-ze kala-rj-ə
boris I.OBJ I.NOM 2SG-INS work-NPST-1SG say-COMP say-PST-3SG
'Boris$_i$ told me that I / *he$_i$ will work with you$_{a(c)}$.'
[Poshkart Chuvash (Turkic), Knyazev 2022: (28)]

Deal proposes that languages with MA agreement display a 'split' in person features: the speaker-denoting feature AUTHOR is divided into two 'sub-persons', AUTH-C and AUTH-I.

(28) **Person features in MA languages (after Deal 2024)**

a. $[\![\text{AUTHOR-C}]\!]^{g,c,i} = \lambda x : s(c) \sqsubseteq x.x$

b. $[\![\text{AUTHOR-I}]\!]^{g,c,i} = \lambda x : s(i) \sqsubseteq x.x$

c. $[\![\text{PARTICIPANT}]\!]^{g,c,i} = \lambda x : s(c) \sqsubseteq x \vee a(c) \sqsubseteq x.x$

Since AUTH-I-specified elements refer to the center of the index, attitudinal quantification will shift its reference to the attitude holder:

(29) **Attitudinal quantification**
$[\![\text{think } \phi]\!]^{g,c} = \lambda x.\forall i'$ compatible with what x thinks in $i : [\![\phi]\!]^{g,c,i'}$

In some languages, AUTH-C and AUTH-I elements come apart, while in others, they are syncretic, being realized uniformly as first person. Languages such as Mishar Tatar or Chuvash in (26) and (27) above illustrate the latter pattern, in which first person agreement on the embedded verb spells out both AUTH-C and AUTH-I features, as does the silent *pro*:

the standard first person, however, spells out only AUTH-C. Logophoric languages such as Donno Sɔ exemplify the former case, in which first person indexicals bear AUTH-C, while LPs are specified with AUTH-I only. Deal (2024) illustrates this with Culy's example in (10), repeated here:

(10) a.

Oumar	inyemɛ	jɛmbɔ	paza	**bolum**	miñ	tagi
Oumar	LOG	sack.DEF	drop	left.1SG	1SG.OBJ	inform.PST

'Oumar$_i$ told me that he$_i$ had left without the sack.'

b.

Oumar	ma	jɛmbɔ	paza	**boli**	miñ	tagi
Oumar	1SG.SBJV	sack.DEF	drop	left.3SG	1SG.OBJ	inform.PST

'Oumar$_i$ told me that I had left without the sack.'

[Donno Sɔ, Culy 1994b: (20)]

In this example, the centers of i and that of c (the speaker) are distinct. Deal (2024) assume that first person agreement in Donno Sɔ realizes AUTH-I, while third person is the elsewhere case. As a consequence, the language has the following agreement probes active:

(30) a. [AUTH-I, F] ⇔ *-um*

b. [F] ⇔ *-i*

Moreover, since Donno Sɔ is a logophoric language, it also makes use of two different pronominal forms, a standard first person, specified with AUTH-C, and a logophoric one, specified with AUTH-I:

(31) a. [AUTH-I, D] ⇔ *inyemɛ*

b. [AUTH-C, D] ⇔ *-ma*

Under a derivational model of Agree in which heads specified with uninterpretable features probe downward to find goals with a (subset of) ϕ-features to satisfy the agreement relation (Chomsky, 2000, 2001; Béjar and Rezac, 2003, 2009; Pesetsky and Torrego, 2007; Preminger, 2014; Deal, 2015, 2020a), the probe looks for a goal that is specified with the same feature. If the controller of the agreement relation is an AUTH-I element – a LP, then the first-person exponent *-um* is inserted: this is the case in (10a), in which the goal is a LP, *inyemɛ*. In the case of (10b), however, the probe is specified with AUTH-I but the goal – a standard first-person pronoun – is specified with AUTH-C: the elsewhere exponent is therefore inserted. Since, in matrix contexts, $s(c) = s(i)$, we correctly predict *-um* to surface with first person in unembedded environments.

An important argument given by Deal (2024) in favor of her treatment of MA (and, by extension, LPs) comes from multiply embedded environments with more than one attitude verb. Deal (2024) observes that, in sentences such as (32), the AUTH-I agreement marker cross refers the center of the most embedded occurence of *gi* 'say', that is, the referent of the second-person pronoun, and not the 'topmost' center, Seydou:

(32)

Se:du	[u	wa	[*pro*	yɔgu	wɔ-ŋ]	gi-y	]	gi-y
Seydou	2SG	QUOT.SBJV		nasty	be-1SG	say-PFV		say-PFV

✓ 'Seydou$_i$ said you said that you are nasty.'

✗ 'Seydou$_i$ said you said that he$_i$ is nasty.'

[Donno Sɔ, Heath 2016, 304]

Deal's analysis predicts this pattern: since AUTH-I-elements denote the center of index i, and that attitudinal quantification quantifies over $s(i)$, AUTH-I-elements in multiply embedded sentences should denote the center of the most local attitude verb. However, as mentioned in §2.1, this prediction seems too strong for most LP-languages, in which no such local dependencies to the closest attitudinal quantifier seems to be at play. In addition to examples (6)-(7) above, the following illustrate the robustness of the data:

(33) Ekpe a-bo ke Udo a-ke a-kere ke imo ì-ke í-kit Ima
Ekpe 3SG-say COMP Udo 3SG-PST 3SG-think COMP LOG LOG-PST LOG-see Ima
'Ekpe$_i$ says that Udo$_j$ thinks that he$_{i/j}$ saw Ima.'
[Ibibio (Niger-Congo: Cross-River), Newkirk 2019, (12)]

(34) Api bO wu ye n kolo n
Api believe COMP LOG likes LOG
'Api$_i$ believes that he$_{i/k}$ likes him$_{i/k}$.'
[Abe (Niger-Congo: Kwa), Koopman and Sportiche 1989, (44a)]

(35) ɱalaŋ yim-go ka: Tulo: ne: ka: yi ŋa mana-m kude.
Malang.M think-PRFVE COMP Tulo.F say COMP LOG PROG.have house-LNK big
'Malang$_i$ thinks that Tuloo$_j$ said that LOG$_{i/j}$ has a big house.'
[Tangale (Afro-Asiatic: Chadic, Nigeria), Haida 2009: (12)]

The problem in this case is that Deal's analysis wrongly predicts the LPs in these examples to only be able to denote the intermediate centers, contrary to fact.

Another potential issue for Deal's account concerns cases in which the overall speaker and the reported speaker are conflated – that is, cases in which $s(c) = s(i)$. In such cases, AUTH-I should be inserted as the exponent of agreement of the verb. As Deal (2024) observes, this is indeed borne out with the 1SG *-N* marker in Donno Sɔ:

(36) mi daːŋa-ŋ gi-y-ŋ
1SG.SBJV sit.STAT-1SG say-PFV-1SG
'I said that I am sitting.'
[Donno Sɔ, Heath 2016, 303]

Assuming that *-N* is syncretic between AUTH-C and AUTH-I features, this is as expected. However, as menttioned in §2.4, LP-languages with dedicated logophoric agreement markers do not behave this way: logophoric agreement/insertion of a logophor tends to be blocked when controllers are first person, (17c)-(18b):

(17) a. aè kɔ aè dɔ-ɛ
3SG said 3SG fell-LOG
'He$_i$ said he$_i$ fell'

b. mm kɔ mm dɔ
1SG said 1SG fell
'I$_i$ said I$_i$ fell'

c. #mm kɔ mm dɔ-ɛ
1SG said 1SG fell-LOG
'I$_i$ said I$_i$ fell' [Gokana, Hyman and Comrie 1981: (2)-(11)]

(18) a. Kofí ŋa bə yi lɔ Áma
Kofi know COMP LOG love Ama
'Kofi$_i$ knows that he$_{i/*j}$ loves Ama'

b. #ŋɟə ŋa bə yi lɔ Áma
1SG know COMP LOG love Ama
Intended: 'I$_i$ know that I$_i$ love Ama'
[Danyi Ewe (Niger-Congo, Togo); O'Neill 2015: (3a, c)]

If logophoric agreement of the kind illustrated here involves an AUTH-I probe, Deal's prediction here is that, regardless of whether the two speakers denote the same (transworld) individuals, if the logophoric agreement marker is an exponent of AUTH-I, then it should be inserted, contrary to fact. A potential solution would be to posit a difference between languages in which agreement is realized uniformly as first person for both AUTH-C and AUTH-I features (as in Donno Sɔ) and languages such as Gokana and Danyi Ewe, which have dedicated logophoric agreement markers (as in Gokana, Danyi Ewe) and therefore, distinct AUTH-C/I probes regulated by a competition principle. Since Deal (2024) does not consider the latter set of languages, she does not provide an explicit answer to this problem.

All in all, in spite of its shortcomings,Deal's analysis highlights a fundamental interpretive connection between first-person and logophoric pronouns, by assuming that they might share a formal feature, AUTH-I, capable of uniformly triggering first person agreement in embedded clauses, a component central to our own account (§4).

3.3 Bassi et al. (2025)

Building on examples such as (21)-(22) illustrating that logophoric pronouns in Ewe, Yoruba and Igbo can systematically eschew binding locality effects under focus, Bassi et al. (2025) propose a system in which logophors are presuppositionally-restricted variables. Their proposal draws on a solution by Sauerland (2013) to account for analogous data involving strict readings of reflexive anaphors, which can exhibit the same interpretive properties in focus-sensitive contexts (McKillen, 2016). Bassi et al. (2025) take this common behavior of LPs and SELF-anaphors as a support that ϕ-features can be ignored during the computation of focus alternatives (Spathas, 2009; Jacobson, 2012; Sauerland, 2013); they assume that LPs are complex elements composed of two different syntactic pieces: a feature LOG, and a variable *pro*. The latter is a variable over individuals concepts (of type $\langle s,e\rangle$) that can either be bound or free, while the former is a presuppositional feature that enforces reference to the attitude holder, ensuring *de se* readings. In Bassi et al. (2025)'s system, which is fully extensional, world variables are present in the syntax and come with every individual or predicate type. A sentence such as (37) has the following truth conditions:

(37) a. Eli$_i$ thinks that LOG$_i$ won.
b. $[\![$(37a)$]\!] = \forall w_x \in Dox_{Eli}, x$ won in w.
c. $[\![$Eli$]\!] = \lambda w_x$. The person in *w* named 'Eli'.
d. $[\![$win$]\!] = \lambda w_x.\lambda z.\ z$ wins in *w*.
e. $[\![\text{LOG}]\!]^g = \lambda f_{\langle s,e\rangle}.\lambda w_x : f(w_x) = x.x$
f. $[\![\text{LOGP}]\!]^g = [\![\text{LOG}]\!]^g([\![pro_i]\!]^g) = [\lambda w_x : [\![pro_i]\!]^g(w_x) = x.x]$

The denotation of LOG is a presuppositional function from world-center pairs to their center, i.e. that individual which the attitude holder takes himself to be in *w*. Since the 'center-mapping' function in which the presupposition of LOG consists of can be ignored during the computation of focus alternatives, this derives strict readings of LPs in both ellipsis and *only*-contexts. Consequently, the account of Bassi et al. (2025) is able to correctly derive most of the interpretations of LPs, including their *de re* readings in sentences involving ellipsis or focus marking due to the presuppositional nature of the LOG feature. However, since for Bassi et al. (2025) an LP is essentially a 3rd person pronoun augmented with LOG, it inherits the problem of HvS concerning the nature and typological status of such a feature. In addition, it fails at explaining the tight connection between logophoricity and first person, as illustrated in §2.3 and §3.2: as opposed to the account of Deal (2024), the fact that person features such as AUTHOR and PARTICIPANT and the LOG feature are defined with respect to the same set of semantic parameters is fully contingent. Last, as it will be discussed in §4.5.2 below, assuming that LPs are internally complex structures composed of [3, LOG] might eventually prove problematic in predicting the disjointness patterns introduced in §2.4.

To summarize, existing analyses all partially fail in predicting the full range of data discussed above: while Deal (2024) accounts for logophoric agreement patterns in assuming that LPs are first-person pronouns, it fails at displaying the required flexibility to license intermediate readings in multiply-embedded clauses, and does not explicity capture the blocking effects observed in cases when logophoric agreement occurs with first person controllers. On the other hand, the account of Bassi et al. (2025), while it significantly improves on Heim-von Stechow's original account by assuming that LPs bear presuppositional features, looses the tight connection between LPs and first-person pronouns required to explain the mismatches in §2.3. As it will become clear in the next section, our own account takes the best of both proposals in assuming that LPs are, at their core, non-indexical pronouns restricted by a presuppositional first-person feature, intergrating them into a full-fledged theory of person features in logophoric marking.

4 Logophors as non-indexical first persons

In this section, we lay out our proposal, which proposes that LPs are, at their core, first-personal elements specified with an +AUTH feature, but – in contrast to genuine first-

Properties	Heim-von Stechow	Deal (2024)	Bassi et al. (2025)
De se readings	✓	✓	✓
No local dependencies	✓	✗	✓
Agreement mismatches	✗	✓	✗
Disjointness inferences	?	?	?

Table 1: Predictions of three theories with respect to key properties of LPs.

person indexicals – additionally specified with a -ACTUAL feature, allowing them to be bound by a subset of attitude verbs in a given language, a proposal inspired by Schlenker (2003).

In what follows, we adopt a Kaplanian two-dimensional semantics in which the interpretation function $[\![\cdot]\!]$ is relativized to a context parameter c, distinct from the index i. We additionally assume an extensional system in which index variables are syntactic objects, and not semantics parameters (Cresswell, 1990; Schlenker, 1999, 2003; Percus, 2000; Keshet, 2008). Following Schwarz (2009, 2012), we furthermore assume that such index pronouns are introduced at the DP level. Crucially for our purposes, we take the index i to be isomorphic to context c (Von Stechow and Zimmermann, 2005; Anand, 2006): both denote tuples consisting of a speaker/center, an (optional) addressee, a world, a time and a location, (38):

(38) **Context-index isomorphism**

a. $c = \langle s(c), a(c), w(c), t(c), l(c)\rangle$

b. $i = \langle s(i), a(i), w(i), t(i), l(i)\rangle$

Depending on the language, some attitude verbs will quantify over indices, returning centered propositions (Lewis, 1979; Chierchia, 1989; Percus and Sauerland, 2003b,a; Anand, 2006; Pearson, 2015), while others merely quantify over propositions Hintikka (1969); Deal (2019). This internal variation regarding modes of quantification is motivated by the observations of Culy (1994a) about logophoric environments mentioned in §2.1 above: for a given language, LPs are only licensed in the scope of a subclass of attitude verbs, following the implicational hierarchy in (5) repeated here:

(5) **A hierarchy of logophoric licensers** [Culy 1994a: (10)]
speech < thought < knowledge < direct perception

This hierarchy tells us that across LP-languages, *say* (and potentially *think*) will be index-quantifiers (39), while a verb such as *know* is more likely to be a world-quantifier only, therefore not licensing LPs, (40):

(39) **Centered attitudinal quantification**

a. $[\![\text{say } \phi]\!]^{g,c} = \lambda i.\lambda x.\forall i' \in \textsc{say}(x,i) : [\![\phi]\!]^{g,c}(i')$
Where $i' \in \textsc{say}(x,i)$ iff what x said in i is compatible with x being $s(i')$ in $w(i')$ at $l(i')$ and $t(i')$.

b. $[\![\text{think } \phi]\!]^{g,c} = \lambda i.\lambda x.\forall i' \in \textsc{dox}(x,i) : [\![\phi]\!]^{g,c}(i')$
Where $i' \in \textsc{dox}(x,i)$ iff x thinks in i that she might be $s(i')$ in $w(i')$ at $loc(i')$ and $t(i')$.

(40) **Attitudinal quantification**
$[\![\text{know } \phi]\!]^{g,c} = \lambda i.\lambda x.\forall w' \in \textsc{know}(x,i) : [\![\phi]\!]^{g,c}(w')$
Where $w' \in \textsc{say}(x,i)$ iff what x knows in i is compatible with what he knows in w'.

Central to the present account (just as in that of Bassi et al. 2025) is that pronouns are, in general, not individuals of type e, but individual concepts, that is, functions from indices to individuals, of type $\langle s, e\rangle$ (Heim, 1998; Büring, 2005; Elbourne, 2008; Von Fintel and Heim, 2011). The pronominal kernel therefore has the following structure, consisting of a pronominal variable *pro* and an index variable i:

(41) **Pronominal kernel**

$$[_{\text{DP}_e}\ [_{\text{NP}}\ pro_{2\,\langle k,e\rangle}]\ \ i_{5\,k}]$$

Both kinds of pronouns (individuals and indices) are interpreted as functions from indices on variables to individuals via the pronouns and trace rule in (42):

(42) **Pronouns and trace rule (Heim and Kratzer, 1998)**
If α is a pronoun or a trace, n is a pronominal index, g an assignment, then

a. $\alpha_n \in dom([\![\cdot]\!]^g)$ iff $n \in dom(g)$;

b. If $\alpha_n \in dom([\![\cdot]\!]^g)$, then $[\![\alpha]\!]^g = g(n)$.

As it is standard, we also assume the following composition rules (from Heim and Kratzer 1998):

(43) **Function application**
If γ is a branching node which daughters are α and β, and if $[\![\alpha]\!]$ is a function whose domain contains $[\![\beta]\!]$, then $[\![\gamma]\!] = [\![\alpha]\!]([\![\beta]\!])$.

(44) **Predicate abstraction**
If α has a binder index λi and β as its daughter constituents, then $[\![\alpha]\!]^g = \lambda x_e.[\![\beta]\!]^{g[i\to x]}$.

4.1 The morphosemantics of person and the ACTUAL feature

The structure in (41) further composes with ϕ-features PERSON, NUMBER and GENDER within a ϕ-phrase (Sauerland, 2003, 2008; Harbour, 2014, 2016); in what follows, we leave out number and gender features, since they are not immediately relevant for the discussion. Crucially, we take person features to be partial functions from individual concepts to individual concepts that introduce presuppositions (Cooper, 1983; Sauerland, 2003, 2008; Heim, 2008; Sudo, 2012): they restrict the value of the individual variable with respect to that of the index i. We assume that the person domain makes use of two primitive features, $\pm$AUTHOR and $\pm$PARTICIPANT (Silverstein, 1976; Noyer, 1992; Harley and Ritter, 2002; McGinnis, 2005; Harbour, 2016), which we abbreviate as $\pm$AUTH and $\pm$PART:

(45) **Person features**

a. $[\![\text{AUTHOR}_{\langle\langle k,e\rangle,\langle k,e\rangle\rangle}]\!]^{g,c} = \lambda P_{\langle k,e\rangle}.\lambda i_k : s(i) \sqsubseteq P(i).P(i)$

b. $[\![\text{PARTICIPANT}_{\langle\langle k,e\rangle,\langle k,e\rangle\rangle}]\!]^{g,c} = \lambda P_{\langle k,e\rangle}.\lambda i_k : s(i) \sqsubseteq P(i) \vee a(i) \sqsubseteq P(i).P(i)$

In their above intensionalized versions, the PART feature denotes a partial function from individual concepts to individual concepts that has to include or be equal to the speaker or addressee in i, while the AUTHOR feature has to include or be equal to the center of i exclusively.[6] These features form a scale based on semantic strenght; AUTH entails PART, but not the other way around.

Following i.a. Noyer (1992), Harbour (2003, 2011b, 2013), Nevins (2007, 2011), and Watanabe (2013), we take person features to be binary,rather than privative. This will have important consequences discussed in §4.5.2, as we will argue that bivalence is required in order to derive the correct distribution of logophoric vs other forms in the paradigms discussed here. However, following Schlenker (2003); Jeretič et al. (2023), we assume that negative features are interpreted trivially: a negatively specified feature will consequently have no semantic import and therefore, no presupposition.

Last, building on the proposal of Schlenker (1999, 2003), an additional $\pm$ACTUAL feature of type $\langle\langle k,e\rangle,\langle k,e\rangle\rangle$, when of positive valence, identifies the value of the index variable i contained in the pronoun with that of c; when negative, it returns the identity function on individual concepts.

(46) **The ACTUAL feature**

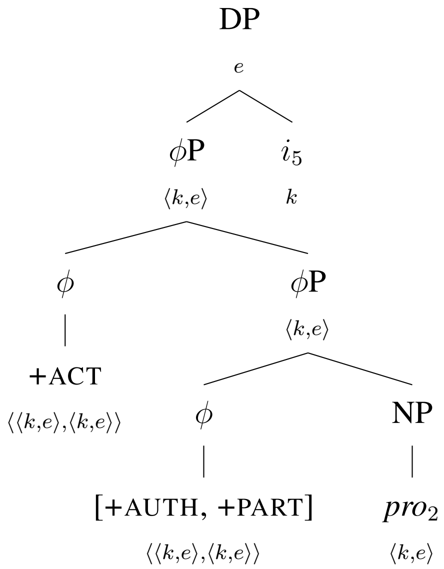
a. $[\![\text{+ACTUAL}_{\langle\langle k,e\rangle,\langle k,e\rangle\rangle}]\!]^{g,c} = \lambda P_{\langle k,e\rangle}\lambda i_k : i = c.P(i)$

b. $[\![\text{-ACTUAL}_{\langle\langle k,e\rangle,\langle k,e\rangle\rangle}]\!]^{g,c} = \lambda P_{\langle k,e\rangle}\lambda i_k.P(i)$

Depending on their functional structure, the valence of $\pm$ACTUAL, and the quantificational force of the attitude verb, pronouns will be interpreted differently in attitude contexts. An indexical pronoun such as the English first person, for instance, will therefore have the structure in (47):

[6] The inclusion relation $\sqsubseteq$ is motivated by the fact that these entries can be pluralized when combined with number features, cf. Sauerland and Bobaljik (2022).

(47) **First-person indexical**



The structure in (47) reflects the fact that in English, first and second person pronouns are indexical and cannot cross-refer the author or addressee of an embedded attitude. Formally, this is captured by the presupposition introduced by +ACT, which ensures that the only possible value for i_5 corresponds to that of c, the utterance context.

(48)
a. $[\![pro_{2_{\langle k,e\rangle}}]\!]^{g,c} = \lambda i.g(2)(i)$

b. $[\![[\text{+AUTH, +PART}]_{\langle\langle k,e\rangle,\langle k,e\rangle\rangle}]\!]^{g,c}([\![pro_2]\!]^{g,c}) =$
$\lambda i.g(2)(i) \text{ iff } \begin{cases} s(i) \sqsubseteq g(2)(i) \\ \text{undefined otherwise} \end{cases}$ *by FA*

c. $[\![\text{+ACT}_{\langle\langle k,e\rangle,\langle k,e\rangle\rangle}]\!]^{g,c}([\![[\text{+AUTH, +PART } [pro_2]]]\!]^{g,c}) =$
$\lambda i.g(2)(i) \text{ iff } \begin{cases} s(i) \sqsubseteq g(2)(i) \wedge i = c \\ \text{undefined otherwise} \end{cases}$ *by FA*

d. $[\![[\text{+ACT } [\text{+AUTH, +PART } [pro_2]]\ i_5]]\!]^{g,c} =$
$g(2)(g(5)) \text{ iff } \begin{cases} s(g(5)) \sqsubseteq g(2)(g(5)) \wedge g(5) = c \\ \text{undefined otherwise} \end{cases}$ *by FA*

The complete pronominal paradigm of a language without LPs such as English can therefore be represented as in (49), with two full-fledged indexical pronouns:

(49) **Featural system without LPs (e.g. English)**

a. 1st: [+AUTH, +PART, +ACTUAL]

b. 2nd: [-AUTH, +PART, +ACTUAL]

c. 3rd: [-AUTH, -PART, -ACTUAL]

When composing with +ACTUAL, pronouns will *not* be affected by (centered) attitudinal quantification: they will be indexical in the Kaplanian sense (Kaplan, 1989). The indexical will always refer to the matrix speaker through identification of i with c, as desired. This is illustrated in (50). It is assumed that i) every predicate projects an index variable, and ii) free index variables (interpreted by the assignment function g) always denote the context of utterance c.

(50) a. Diane thought that I was funny.

b. $[\![\text{funny}_{\langle k\langle e,t\rangle\rangle}\ i_3]\!]^{g,c} = \lambda x.x$ is funny in $g(3)$

c. $[\![\text{I}]\!]^{g,c} = g(2)(g(5))\text{iff} \begin{cases} s(g(5)) \sqsubseteq g(2)(g(5)) \wedge g(5) = c \\ \text{undefined otherwise} \end{cases}$ $\sim s(c)$

d. $[\![\text{funny } i_3]\!]^{g,c}([\![\text{I}]\!]^{g,c}) = s(c)$ is funny in $g(3)$ *by FA*

e. $[\![\lambda i_3 \text{ [I [funny}_{i_3}]]]\!]^{g,c} = \lambda i_3.s(c)$ is funny in i_3 *by PA*

f. $[\![\text{think}_{\langle\langle k,t\rangle\langle e\langle k,t\rangle\rangle\rangle}\ i_7]\!]^{g,c} = \lambda p_{\langle k,t\rangle}.\lambda x_e.\forall i'$ compatible with what x thinks in $g(7)$, $p(i')$

g. $[\![\text{think}_{i_7}]\!]^{g,c}([\![\lambda i_3 \text{ [I [funny}_{i_3}]]]\!]^{g,c}) = \lambda x.\forall i'$ compatible with what x thinks in $g(7)$, $s(c)$ is funny in i' *by FA*

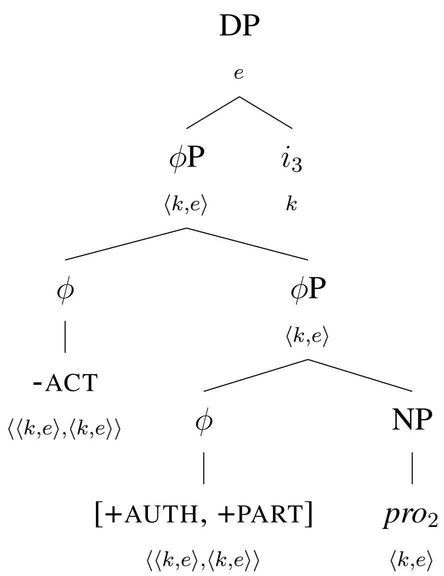
h. $[\![\text{[think}_{i_7} \text{ [}\lambda i_3 \text{ [I [funny}_{i_3}]]]]]\!]^{g,c}([\![\text{Diane}]\!]^{g,c}) = 1$ iff $\forall i'$ compatible with what Diane thinks in $g(7)$, $s(c)$ is funny in i' *by FA*

We ensure that the (indexical) presupposition projects to the matrix clause by adopting Heim (1992)'s presupposition projection (PP) rule in (51):

(51) **Presupposition projection under attitudes (Heim, 1992)**
$[\![x \text{ says } p]\!]^{g,c,i}$ is defined iff $\forall i' \in \text{SAY}(x,i) : [\![p]\!]^{g,c,i'}$ is defined.

By contrast, if a given language lexicalizes the -ACTUAL feature, it will have a 'center-denoting' first person that need not be the actual center in its paradigm - that is, a logophoric pronoun. Logophoric paradigms thus share a common basis with English-like ones, with the important difference that in addition to 'genuine' first person forms, they also make available one (or more; see §4.3) additional pronominal structures specified with -ACTUAL. As a consequence, the index variable contained in the pronoun is not required to be identified with the context of utterance, and is able to be bound by the attitude verb. Under this view, a LP is therefore a strictly non-indexical first person, much like AUTH-I-specified elements in the proposal of Deal (2024) (cp. also Schlenker 1999, 2003). Consequently, a center(speaker)-denoting LP has the structure in (52):

(52) **First-person logophoric pronoun**



(53) a. $[\![pro_{2_{\langle k,e\rangle}}]\!]^{g,c} = \lambda i.g(2)(i)$

b. $[\![[\text{+AUTH, +PART}]_{\langle\langle k,e\rangle,\langle k,e\rangle\rangle}]\!]^{g,c}([\![pro_2]\!]^{g,c}) =$
$\lambda i.g(2)(i) \text{ iff } \begin{cases} s(i) \sqsubseteq g(2)(i) \\ \text{undefined otherwise} \end{cases}$ *by FA*

c. $[\![\text{-ACT}_{\langle\langle k,e\rangle,\langle k,e\rangle\rangle}]\!]^{g,c}([\![[\text{+AUTH, +PART } [pro_2]]]\!]^{g,c}) =$
$\lambda i.g(2)(i) \text{ iff } \begin{cases} s(i) \sqsubseteq g(2)(i) \\ \text{undefined otherwise} \end{cases}$ *by FA*

d. $[\![[\text{-ACT } [\text{+AUTH, +PART } [pro_2]]\ i_3]]\!]^{g,c} =$
$g(2)(g(3)) \text{ iff } \begin{cases} s(g(3)) \sqsubseteq g(2)(g(3)) \\ \text{undefined otherwise} \end{cases}$ *by FA*

Since the pronominal complex is -ACT, the index pronoun i_3 it contains can either be identified throught the assignment function g to the actual context c, or be bound by the binder introduced by the attitude verb: this is demonstrated in the following derivation, with the LF in (54a):

(54) a. Diane thought that λi_3 [LOG$_2$ [i_3]] was funny.

b. $[\![\text{funny}_{\langle k\langle e,t\rangle\rangle}\ i_3]\!]^{g,c} = \lambda x.x$ is funny in $g(3)$

c. $[\![[\text{LOG}_2\ [i_3]]]\!]^{g,c} = g(2)(g(3)) \text{ iff } \begin{cases} s(g(3)) \sqsubseteq g(2)(g(3)) \\ \text{undefined otherwise} \end{cases}$
$\sim s(g(3))$

d. $[\![\text{funny } i_3]\!]^{g,c}([\![[\text{LOG}_2\ [i_3]]]\!]^{g,c}) = s(g(3))$ is funny in $g(3)$ *by FA*

e. $[\![\lambda i_3\ [[\text{LOG}_2\ [i_3]]\ [\text{funny}_{i_3}]]]\!]^{g,c} = \lambda i_3.s(i_3)$ is funny in i_3 *by PA*

f. $[\![\text{think}_{\langle\langle k,t\rangle\langle e\langle k,t\rangle\rangle\rangle}\ i_7]\!]^{g,c} = \lambda p_{\langle k,t\rangle}.\lambda x_e.\forall i'$ compatible with what x thinks in $g(7)$, $p(i')$

g. $[\![\text{think}_{i_7}]\!]^{g,c}([\![\lambda i_3\ [\text{LOG}_{i_3}\ [\text{funny}_{i_3}]]]\!]^{g,c}) = \lambda x.\forall i'$ compatible with what x thinks in $g(7)$, $s(i')$ is funny in i' *by FA*

h. $[\![[\text{think}_{i_7}\ [\lambda i_3\ [\text{LOG}_{i_3}\ [\text{funny}_{i_3}]]]]]\!]^{g,c}([\![\text{Diane}]\!]^{g,c}) = 1$ iff $\forall i'$ compatible with what Diane thinks in $g(7)$, $s(i')$ is funny in i' *by FA*

The full paradigm of a language with speaker logophors such as Ewe would therefore be represented as follows:

(55) **Featural system with speaker logophor (e.g. Ewe)**

a. 1st: [+AUTH, +PART, +ACTUAL]

b. LOG: [+AUTH, +PART, -ACTUAL]

c. 2nd: [-AUTH, +PART, +ACTUAL]

d. 3rd: [-AUTH, -PART, -ACTUAL]

An immediate consequence of the present proposal is that, like its predecessors, it predicts the default *de se* readings of LPs mentioned in §2.2; since LPs are specified with a first-person feature +AUTH, they will always denote the center of the index i, as desired; the *de re* readings of LPs attested in Ewe can be derived using concept generators (Percus and Sauerland, 2003a), as in the original proposal of Pearson (2015), to which we refer the reader for more details. Another consequence is that, since LPs are individual concepts that contain, at their core, an index variable i, intermediate readings of the kind discussed in §2.1 in cases in which LPs are embedded under more than one attitude verb are now predicted. Recall that, in languages such as Ewe, Yoruba and Ibibio, among many others, LPs can obtain their reference at the level of both speech acts:

(6) a. Kofi xɔ-e se be Ama gblɔ be yè-ʃu yè.
Kofi receive hear COMP Ama say COMP LOG-beat LOG
'Kofi$_i$ believed that Ama$_j$ said that LOG$_{i/j}$ beat LOG$_{i/j}$.'
[Ewe, Clements 1975, 73]

b. Marie be Kofi xɔse be yè na yè cadeau.
Mary say Kofi believe COMP LOG give LOG gift
'Mary$_i$ said that kofi$_j$ believed that LOG$_{i/j}$ gave LOG$_{i/j}$ a gift.'
[Ewe, Pearson 2015, (47)]

(33) Ekpe a-bo ke Udo a-ke a-kere ke imo ì-ke í-kit Ima
Ekpe 3SG-say COMP Udo 3SG-PST 3SG-think COMP LOG LOG-PST LOG-see Ima
'Ekpe$_i$ says that Udo$_j$ thinks that he$_{i/j}$ saw Ima.'
[Ibibio, Newkirk 2019, (12)]

Now our system is flexible enough to account for it: the index variable of LOG can be bound by the local abstractor (as above) or the intermediate one, as a consequence of the index variable being syntactically realized (Percus, 2000; Schwarz, 2012). The derivation in (56) illustrates the former case, in which the index variable of LOG is bound by the most local λ-binder introduced by *think*:

(56) a. Ekpe says that [λi_7 Udo thinks that [λi_5 [LOG$_2$ [i_5]] saw Ima]]

b. $[\![$saw Ima$_{\langle k\langle e,t\rangle\rangle}$ $i_5]\!]^{g,c} = \lambda x.x$ saw Ima in $g(5)$

c. $[\![$saw Ima $i_5]\!]^{g,c}([\![[$LOG$_2$ $[i_5]]]\!]^{g,c}) = s(g(5))$ saw Ima in $g(5)$ *by FA*

d. $[\![\lambda i_5$ [[LOG$_2$ $[i_5]]$ [saw Ima$_{i_5}$]]$]\!]^{g,c} = \lambda i_5.s(i_5)$ saw Ima in i_5 *by PA*

e. $[\![$think$_{\langle\langle k,t\rangle\langle e\langle k,t\rangle\rangle\rangle}$ $i_7]\!]^{g,c} = \lambda p_{\langle k,t\rangle}.\lambda x_e.\forall i'$ compatible with what x thinks in $g(7)$, $p(i')$

f. $[\![$think$_{i_7}]\!]^{g,c}([\![\lambda i_5$ [[LOG$_2$ $[i_5]]$ [saw Ima$_{i_5}$]]$]\!]^{g,c}) = \lambda x.\forall i'$ compatible with what x thinks in $g(7)$, $s(i')$ saw Ima in i' *by FA*

g. $[\![$[think$_{i_7}$ [λi_5 [LOG$_{i_5}$ [saw Ima$_{i_5}$]]]]$]\!]^{g,c}([\![$Udo$]\!]^{g,c}) = 1$ iff $\forall i'$ compatible with what Udo thinks in $g(7)$, $s(i')$ saw Ima in i' *by FA*

h. $[\![\lambda i_7$ [Udo [think$_{i_7}$ [λi_5 [LOG$_{i_5}$ [saw Ima$_{i_5}$]]]]]$]\!]^{g,c}) = \lambda i_7.\lambda x.\forall i'$ compatible with what x thinks in i_7, $s(i')$ saw Ima in i' *by PA*

i. $[\![$say$_{\langle\langle k,t\rangle\langle e\langle k,t\rangle\rangle\rangle}$ $i_8]\!]^{g,c} = \lambda p_{\langle k,t\rangle}.\lambda x_e.\forall i'$ compatible with what x said in $g(8)$, $p(i')$

j. $[\![$say$_{i_8}]\!]^{g,c}([\![\lambda i_7$ [Udo [think$_{i_7}$ [λi_5 [LOG$_{i_5}$ [saw Ima$_{i_5}$]]]]]$]\!]^{g,c}) = \lambda x.\forall i'$ compatible with what x said in $g(8)$, $\forall i''$ compatible with what Udo thinks in i', $s(i'')$

thinks that $s(i'')$ saw Ima in i'' *by FA*

k. $[\![$[say$_{i_8}$ [λi_7 [Udo [think$_{i_7}$ [λi_5 [LOG$_{i_5}$ [saw Ima$_{i_5}$]]]]]]]]$]\!]^{g,c}$($[\![$Ekpe$]\!]^{g,c}$) $= \forall i'$ compatible with what Ekpe said in $g(8)$, $\forall i''$ compatible with what Udo thinks in i', $s(i'')$ saw Ima in i'' *by FA*

The above derivation therefore delivers the most local interpretation of LOG, in which it is bound by the λ-binder introduced by *think* and denotes the center of the worlds that it quantifies over, i.e., Udo.

The second reading of (33), in which the LP refers to the matrix subject *Ekpe*, is captured by letting the pronoun being bound by the higher lambda-binder λi_7 introduced by *say*:

(57) a. Ekpe says that [λi_7 Udo thinks that [λi_5 [LOG$_2$ [i_7]] saw Ima]]

b. $[\![$saw Ima$_{\langle k\langle e,t\rangle\rangle}$ $i_7]\!]^{g,c} = \lambda x.x$ saw Ima in $g(7)$

c. $[\![$saw Ima $i_7]\!]^{g,c}$($[\![$[LOG$_2$ [i_7]]$]\!]^{g,c}$) $= s(g(7))$ saw Ima in $g(7)$ *by FA*

d. $[\![\lambda i_5$ [[LOG$_2$ [i_7]] [saw Ima$_{i_5}$]]$]\!]^{g,c} = \lambda i_5.s(g(7))$ saw Ima in i_5 *by PA*

e. $[\![$think$_{\langle\langle k,t\rangle\langle e\langle k,t\rangle\rangle\rangle}$ $i_7]\!]^{g,c} = \lambda p_{\langle k,t\rangle}.\lambda x_e.\forall i'$ compatible with what x thinks in $g(7)$, $p(i')$

f. $[\![$think$_{i_7}]\!]^{g,c}$($[\![\lambda i_5$ [[LOG$_2$ [i_7]] [saw Ima$_{i_5}$]]$]\!]^{g,c}$) $= \lambda x.\forall i'$ compatible with what x thinks in $g(7)$, $s(g(7))$ saw Ima in i' *by FA*

g. $[\![$[think$_{i_7}$ [λi_5 [LOG$_{i_7}$ [saw Ima$_{i_5}$]]]]$]\!]^{g,c}$($[\![$Udo$]\!]^{g,c}$) $= 1$ iff $\forall i'$ compatible with what Udo thinks in $g(7)$, $s(g(7))$ saw Ima in i' *by FA*

h. $[\![\lambda i_7$ [Udo [think$_{i_7}$ [λi_5 [LOG$_{i_7}$ [saw Ima$_{i_5}$]]]]]$]\!]^{g,c}$) $= \lambda i_7.\lambda x.\forall i'$ compatible with what x thinks in i_7, $s(i')$ saw Ima in i' *by PA*

i. $[\![$say$_{\langle\langle k,t\rangle\langle e\langle k,t\rangle\rangle\rangle}$ $i_8]\!]^{g,c} = \lambda p_{\langle k,t\rangle}.\lambda x_e.\forall i'$ compatible with what x said in $g(8)$, $p(i')$

j. $[\![$say$_{i_8}]\!]^{g,c}$($[\![\lambda i_7$ [Udo [think$_{i_7}$ [λi_5 [LOG$_{i_7}$ [saw Ima$_{i_5}$]]]]]$]\!]^{g,c}$) $= \lambda x.\forall i'$ compatible with what x said in $g(8)$, $\forall i''$ compatible with what Udo thinks in i', $s(i')$ saw Ima in i'' *by FA*

k. $[\![$[say$_{i_8}$ [λi_7 [Udo [think$_{i_7}$ [λi_5 [LOG$_{i_7}$ [saw Ima$_{i_5}$]]]]]]]]$]\!]^{g,c}$($[\![$Ekpe$]\!]^{g,c}$) $= \forall i'$ compatible with what Ekpe said in $g(8)$, $\forall i''$ compatible with what Udo thinks in i', $s(i')$ saw Ima in i'' *by FA*

On this second, 'intermediate' reading, the LP is bound by the lambda-binder λi_7 introduced by the topmost predicate *say*, thus constraining the pronominal index of LOG to denote the center of the worlds quantified over by *say* - that is, Ekpe.[7] Similar reasoning applies *ceteris paribus* to (7) and (35).

[7]Note that, in the above derivations, we assumed that the index introduced by the predicate *saw Ima* and that introduced by LOG were systematically (un)bound together by the same λ. This is not a requirement of the theory, and nothing in the system prevents the predicate *saw Ima* to be bound by another binder, and therefore not interpreted with respect to the same set of worlds as LOG. Such binding restrictions are expected, however: see i.a. Percus (2000); Keshet (2010); Schwarz (2012).

4.2 Deriving agreement mismatches

In the present system, just as in Deal (2024), LPs are essentially non-indexical, first-personal elements. It is worth stressing that this theoretical claim is bolstered by morphological regularities observable in the languages at stake. Indeed, the first-personal nature of LPs as an hypothesis to explain their distribution and evolution is not new (Westermann 1907; Clements 1975); it was invoked notably by Faltz (1985) to explain, among other things, the fact that in the Anlo dialect of Ewe, the first person form *ye* is used both as the first-person logophor and as the first person singular genitive clitic in matrix clauses (Faltz 1985; pp. 261 *sqq*), suggesting a common first-personal origin. This morphological overlap is illustrated in Table 2 for Ewe, Table 3 for Ibibio, and Table 4 for Wan, respectively; in the Ewe and Ibibio paradigms, the morphemes corresponding to the first person reappear in the logophoric forms in both the singular and plural, while the syncretism is transparent in Wan only in the plural. This is the case for standard pronominal forms in all three languages, as well as for subject and object agreement markers in Ibibio.

Person	Pronouns	Subject clitics	Emphatic/Possessive pronouns
1SG	-m	me-	nye
2SG	wò	(n)è-	wò
3SG	-í	wò/é	é
1PL	mí	mí(é)	miá(wó)
2PL	mí	mi(e)	mia(wó)
3PL	wó	wó	wó
LOG.SG	ye	ye	ye
LOG.PL	yewó	yewó	yewó

Table 2: Ewe pronouns (Ameka, 1991, 56)

Person	Subject pronouns	Object pronouns	Subject agreement	Object agreement
1SG	ami	mien	n-	n-
2SG	afo	fien	a-	u-
3SG	anye	anye	a-	a-
1PL	nnyin	nnyin	i-	i-
2PL	ndufo	ndufo	e-	e-
3PL	ommo	ommo	e-	e-
LOG.SG	imo	imo	i-	i-
LOG.PL	mmimo	mmimo	i-	i-

Table 3: Ibibio pronouns (Newkirk, 2019, 310)

Such syncretisms between the first person exclusive marker and the logophoric form strongly suggests that both share an inherent first person feature. This vindicates an approach of logophoric elements being inhrently first-person, *contra* alternatives analyses such as that of Heim-von Stechow and Bassi et al. (2025), which take logophors to be third person pronons augmented with a LOG feature (see §4.5.2 below). In addition, the proposal of LPs being inherently first personal straightforwardly explains the apparent 'mismatches' introduced in §2.3 for some languages such as Donno Sɔ, Ibibio, Ekpeye and Efik, in which LPs trigger first person agreement on the embedded verb:

Person	Independent form	Subject pronouns	Object pronouns
1SG	amĩŋ, mĩnã	ǹ, nã	ǹ
2SG	ɓíla	la	la
3SG	è	è	à
1PL.DU	kó	kó	ko
1PL.EXCL	kã	kã	kã
1PL.INCL	kà	kà	kà
2PL	ãŋ	ãŋ	ãŋ
3PL	ãŋ	ãŋ	ãŋ
LOG.SG	ɓã	ɓã	ɓã
LOG.PL	mɔŋ	mɔŋ	mɔŋ

Table 4: Wan pronouns (after Ravenhill 1974; Nikitina 2012a, 2023)

(2) a. Oumar inyemɛ jɛmbɔ paza **bolum** miñ tagi
Oumar LOG sack.DEF drop left.1SG 1SG.OBJ inform.PST
'Oumar$_i$ told me that he$_i$ had left without the sack.'

b. Oumar ma jɛmbɔ paza **boli** miñ tagi
Oumar 1SG.SBJV sack.DEF drop left.3SG 1SG.OBJ inform.PST
'Oumar$_i$ told me that I had left without the sack.'

[Culy 1994b: (20)]

(11) a. Ekpe a-bo ke imo ì-ma í-to Udo
Eke 3SG.say COMP LOG LOG-PST LOG-hit Udo
'Ekpe$_i$ says that he$_i$ hit Udo.'

b. ommo e-ke e-bo ke mmimo ì-ma ì-kot nwet
3PL 3PL-PST 3PL-say COMP LOG.PL 1PL-PST 1PL-read book
'They$_i$ said that they$_i$ read a book.'

[Ibibio, adapted from Newkirk 2019: (1a)-(5)]

Under the view that agreement requires feature sharing or subsumption (Frampton and Gutmann, 2000; Pesetsky and Torrego, 2007; Ackema and Neeleman, 2013), the Donno Sɔ and Ibibio patterns follows straightforwardly: speaker logophors bearing an AUTH feature, the presence of first person agreement on the target embedded verb is expected, reflecting the features of their controller. The Ibibio pattern is captured by positing that agreement markers are syncretic between first person plurals and logophors:[8]

(58) a. [+AUTHOR, +PART, +ACTUAL, +SG] ⇔ /n-/

b. [+AUTHOR, +PART, +ACTUAL, -SG] ⇔ /i-/

c. [+AUTHOR, +PART, -ACTUAL, ±SG] ⇔ /i-/

The Subset Principle of Halle (1997) will consequently ensure that, in a language such as Ibibio, logophoric marking (and not first-person marking) appears regardless of number in cases in which the features of its controller contains -ACTUAL:

[8] Arguably, the label ±SG for representing number features is incorrect: see Harbour 2007, 2011b, 2014.

(59) **Subset Principle (Halle, 1997, 428)**

a. The phonological exponent of a Vocabulary Item is inserted into a position if the item matches all or a subset of the features specified in the terminal morpheme;

b. Insertion does not take place if the Vocabulary Item contains features not present in the morpheme;

c. Where several Vocabulary Items meet the conditions for insertion, the item matching the greatest number of features specified in the terminal morpheme must be chosen.

We can also account for languages such as Karimojong (in which apparent third person triggers first person agreement in logophoric contexts) by assuming that, in such a language, the third person pronoun is syncretic with a [+AUTH, -ACT] element – a logophor – and that the probe is insensitive to the valence of ACT, allowing insertion of the first person agreement marker with first person pronouns in matrix environments:

(13)

àbu	papà	tolim	ɛbè	àlózì	iŋèz	morotó
AUX	father	say	COMP	1SG.go.NPST	3SG	Moroto

'Father$_i$ said that he$_i$ was going to Moroto.'

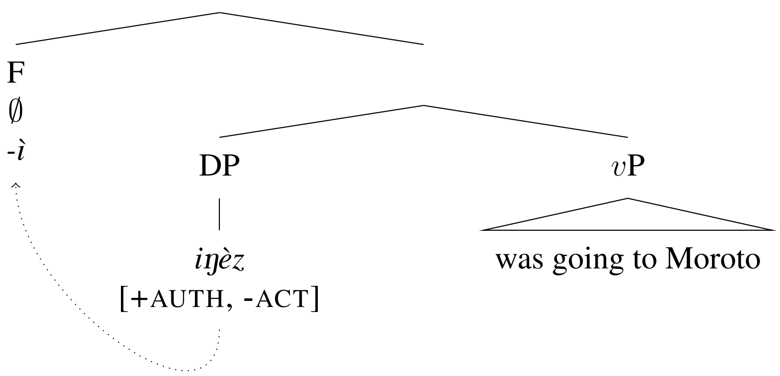
[Karimojong, Curnow 2002b: (18)]

(60) a. [+AUTH, +PART -ACT, D] ↔ *iŋèz*

b. [-AUTH, -PART -ACT, D] ↔ *iŋèz*

c. [+AUTH, +PART, ±ACT, F] ↔ *ì*

(61) **Derivation of (13)**



The case of Ekpeye, which shows an alternation between a dedicated logophoric marker in the singular and a syncretic first person/logophoric marker in the plural is more complex:

(12) a. ù-kà ɓú yá’ zè
3SG-say.PST COMP.NON1 LOG.SG go.PST
‘He$_i$ said that he$_i$ went.’

b. ù-kà-ɓɛ ɓú à-zè
3SG-say.PST-PL COMP.NON1 1PL-go.PST
‘They$_i$ said that they$_i$ went.’
[Ekpeye, adapted Curnow 2002b: (27)-(29), after Clark 1972]

A solution presents itselfs if we assume that Ekpeye agreement morphemes differ with respect to number: the singular logophoric agreement morpheme has a distinct exponent from the first person (which seems to be zero marking), while in the plural logophoric and first person exponents are syncretic.

(62) a. [+AUTHOR, +PART, -ACTUAL, +SG] ⇔ ∅

b. [+AUTHOR, +PART, ±ACTUAL, -SG] ⇔ /à-/

Additionally, our system (just like that of Deal 2024) can be extended to languages displaying monstrous agreeement, as discussed in §3.2: under our account, monstrous agreeement is similar to logophoric agreement, except that there is no agreement exponence specific to the bundle [+AUTH, -ACT, F]; probes are only sensitive to [+AUTH], just like in Karimojong.

(27) a. boris man-a ***pro*** san-ba ëc̨l-e-p te-ze kala-rj-ə
boris I.OBJ ***pro*** 2SG-INS work-NPST-1SG say-COMP say-PST-3SG
‘Boris$_i$ told me that I / he$_i$ will work with you$_{a(c)}$.’

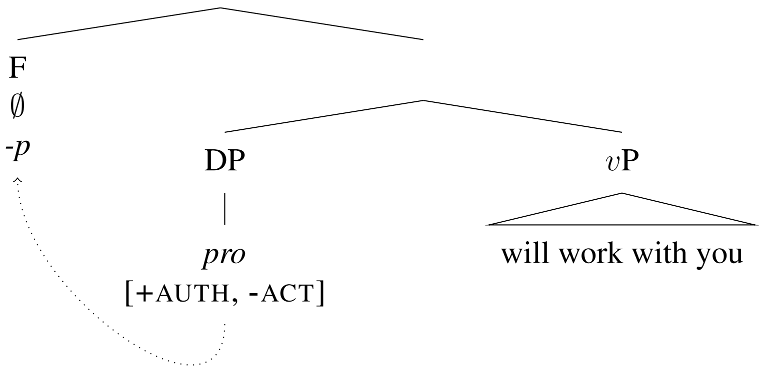
b. boris man-a ep san-ba ëc̨l-e-p te-ze kala-rj-ə
boris I.OBJ I.NOM 2SG-INS work-NPST-1SG say-COMP say-PST-3SG
‘Boris$_i$ told me that I / *he$_i$ will work with you$_{a(c)}$.’
[Poshkart Chuvash (Turkic), Knyazev 2022: (28)]

(63) a. [+AUTH, ±ACT, F] ⇔ *-p*

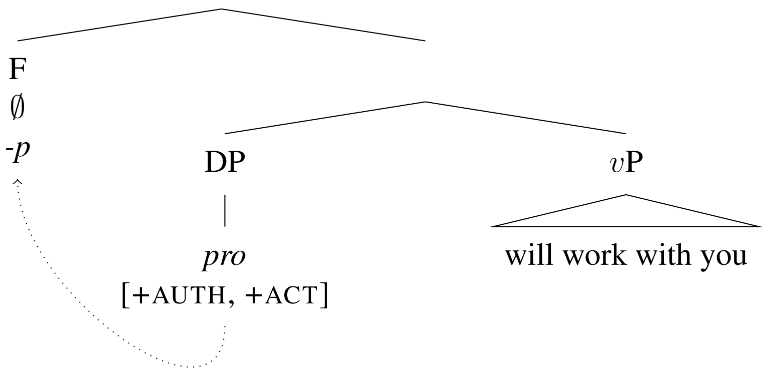
b. [+AUTH, ±ACT, D] ⇔ *pro*

c. [+AUTH, +ACT, D] ⇔ *-ep*

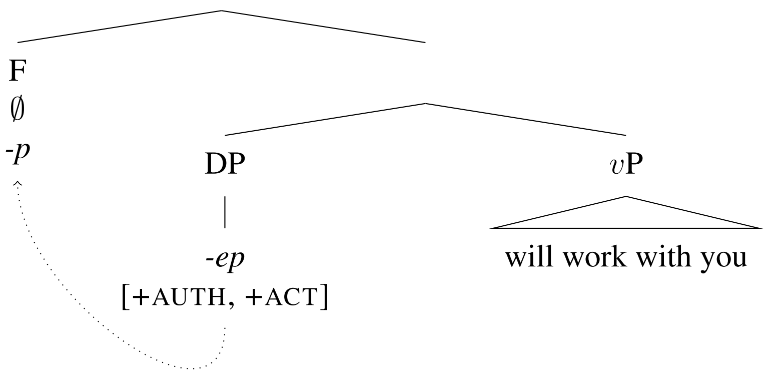
(64) a. **Derivation of (27a) (monstrous reading)**



b. **Derivation of (27a) (indexical reading)**



c. **Derivation of (27b)**



4.3 Parametrization of ACTUAL

Another advantage of the present approach is that, ±ACT being a feature independent of person, it can compose in different ways inside the pronominal structure to derive various combinations of person logophors attested in natural language. Put differently, if the feature ACTUAL is able to lexicalize on different persons, then we would expect to find languages that realize [+AUTH, +PART, -ACT] and/or [-AUTH, +PART, -ACT] – that is, first- and second-person logophors. This is indeed borne out: languages Goemai and Mupun (West Chadic, Nigeria), for instance, have both LOG authors and LOG addressees:

(65)

k'wal	yin	gwa	goe	tu	ji
talk	say	SG.M.LOG.2	OBLIG	kill	SG.M.LOG.1

'He$_i$ said he$_j$ should kill him$_i$'
(lit. 'He$_i$ said you$_j$ should kill me$_i$')

[Goemai, Hellwig 2006: 219]

(66)

n-sat	n-wur	nə	**gwar**	ji
1SG-say	PREP-3SG	COMP	**2SG.LOG**	come

'I told him$_i$ that he$_i$ should come'
(lit. 'I told him that 2SG.LOG should come.')

[Mupun, Frajzyngier 1997: (35)]

Both paradigms are adequatly captured with the person hierarchy in (67):

(67) **Featural system with speaker and addressee LOGs**

a. 1st: [+AUTH, +PART, +ACTUAL]

b. LOG1: [+AUTH, +PART, -ACTUAL]

c. LOG2: [-AUTH, +PART, -ACTUAL]

d. 2nd: [-AUTH, +PART, +ACTUAL]

e. 3rd: [-AUTH, -PART, -ACTUAL]

Note that it is predicted that in such a paradigm, the second person can only be used in logophoric contexts to denote actual addressees, as example (68) confirms: the 2SG form *wur* can only refer to the actual addressee, $a(c)$.

(68)

n-sat	n-wur	nə	**wur**	ji
1SG-say	PREP-3SG	COMP	**2SG**	come

'I told him that you$_{a(c)}$ should come.'

[Mupun, Frajzyngier 1997: (36)]

Since no restriction is placed on the lexicalization of ACT with respect to other features, we expect to find languages with LOG addressees, but no LOG authors. This is the case of West Chadic language Pero:

(69)

ca	peemu	ta	kayu	laa	mu	mijiba
say.PST	LOG.2SG	FUT	drive away	man	DEM	stranger

'[He] said that he$_{a(c')}$ is going to drive the stranger away.'
(lit. '[He] said that you$_{a(c')}$ are going to drive the stranger away.')

[Frajzyngier 1985: (23b)]

The Pero pattern can be described with the following hierarchy, in which the bundle [+AUTH, +PART, -ACTUAL] is not lexicalized.

(70) **Featural system with addressee LOG only**

a. 1st: [+AUTH, +PART, +ACTUAL]

b. LOG2: [-AUTH, +PART, -ACTUAL]

c. 2nd: [-AUTH, +PART, +ACTUAL]

d. 3rd: [-AUTH, -PART, -ACTUAL]

Typologically, paradigms described in (67) and (70) are rare, only to be found in languages pertaining to the Chadic branch (Nikitina, 2012b).

4.3.1 The other PARTICIPANT

In the present theory is correct about the distribution of ACT, then it should be possible to find paradigms converse to (67) or (70), that is, paradigms in which first and/or second persons are not restricted to actual participants. This is the case of Wan (Niger-Congo, Ivory Coast), which allows second person pronouns to refer to reported addressees, in addition to actual ones:

(71)

è	gé	zò	ɓé	**là**	**ɓà**	pólì
3SG	said	come	then	2SG	LOG.SG	wash

'She$_i$ said come and wash me$_i$.' [Wan, Nikitina 2012a: (18)]

In (71), the second person pronoun *là* refers to the addressee of the reported speech act, $a(i)$, and not to the current addressee: in other words, the second person in Wan is logophoric. Analogous patterns can be found for the languages Aghem (Hyman and Watters 1979; Butler 2009), Mundang (Hagège, 1974), Engenni (Thomas, 1978), and Akoose (Hedinger, 1984) (all Niger-Congo). The hierarchy in (72) captures this pattern, in which the second person pronoun is specified with -ACT:

(72) **Featural system with speaker LOG and -ACTUAL 2P (Wan, Aghem)**

a. 1st: [+AUTH, +PART, +ACT]

b. LOG: [+AUTH, +PART, -ACT]

c. 2nd: [-AUTH, +PART, -ACT]

d. 3rd: [-AUTH, -PART, -ACT]

However, in some other languages, including Ewe and Donno Sɔ, second person marking always refer to actual addressees and cannot be used for reported addressees - third person must be used in that case. This is illustrated in (73)-(74):

(73)

Be	indvembe	velaa	uñ	tembeliñ	giya
3PL	LOG.PL	come	2SG.OBJ	found.NEG.1PL	said.3PL

'They$_i$ said that they$_i$ didn't find you when they$_i$ came.'
[Culy 1994b: (6b), after Kervran and Prost 1986]

(74) Kofi gblɔ na wo bè yè-a-dyi ga-a na wo
Kofi speak to 3PL COMP LOG-T-seek money-D for 3PL
'Kofi$_i$ said to them$_j$ that he$_i$ would seek the money for them$_j$.'
[Nikitina 2012a: (23), after Clements 1975]

In (73), the second-person object is indexical and cannot refer to the reported addresse; such reference can only be achieved by using third person, as in English or Ewe, which (74) illustrates. This seems to suggest that languages such as Wan and Ewe differ in the valence of ±ACTUAL feature on the second person, restricting or not its referent to the current speech act participants:

(75) **Featural system with speaker LOG and +ACTUAL 2P (Ewe, Donno Sɔ)**

a. 1st: [+AUTH, +PART, +ACT]

b. LOG: [+AUTH, +PART, -ACT]

c. 2nd: [-AUTH, +PART, +ACT]

d. 3rd: [-AUTH, -PART, -ACT]

As a consequence, languages with systems such as (75) possess two 'genuine' indexical forms with different person specifications, alongside a full-fledged LOG form.

In light of the above, it is relevant to compare our proposal to that of Deal (2024), which posits an independent person feature, AUTH-I, to account for both monstrous and logophoric agreement (§3.2). Capturing the above data in Deal's system would require to posit another person feature, PART-I, since it is commonly accepted that second persons do not bear (positively-valued) AUTHOR features (Noyer, 1992; Harley and Ritter, 2002; McGinnis, 2005; Nevins, 2007; Watanabe, 2013; Harbour, 2016). By contrast, our take on the person inventories that logophoric systems make use of is conservative: what distinguishes second person pronouns from addressee-denoting LPs is the valence of ACT, not a different person specification. In addition of being more parsimonious, the present analysis therefore allows us to explain the difference between Ewe and Wan systems without systematically appealing to syncretism: the reason why second persons in Wan, Aghem and many others can denote non-actual participants is because they bear -ACT, not because PART-C and PART-I are systematically syncretic in those languages.

Present system	**Deal (2024)**
±AUTH	AUTH-C
±PART	AUTH-I
±ACT	PART-C
	PART-I

Table 5: Featural ontologies of the present system vs. that of Deal (2024).

4.4 Excursus: shifted indexicals

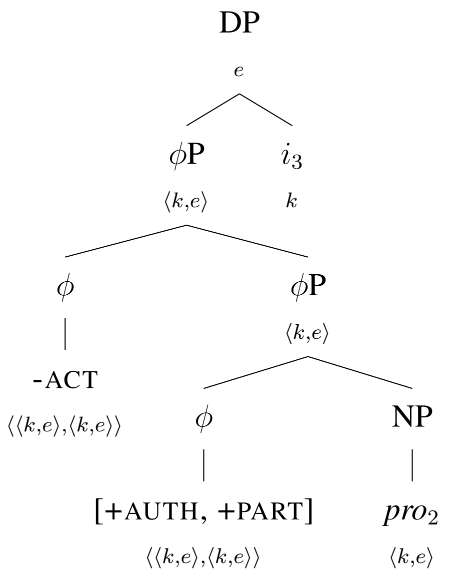
In some languages, first and second person pronouns can receive a logophoric interpretation in complex finite clauses. This is illustrated in (76) for Zazaki (Indo-Iranian, Eastern Turkey):

(76) Hɛsen-i mɨ-ra va kɛ **ɛz** dɛwletia
Hesen-OBL 1SG-OBL say COMP **1SG.NOM** rich.be.PRS
'Hesen$_i$ tells me$_{s(c)}$ that he$_{i/s(c)}$ is rich.'
[Zazaki, Anand and Nevins 2004: (4)]

In (76), the nominative first person *ɛz* embedded under *va* '*say*' can either refer to Hesen or the utterance speaker. Such 'shifted indexicals' been reported for languages pertaining to different, typologically unrelated families, ranging from Semitic (Amharic, Schlenker 1999, 2003, LaTerza et al. 2015; Ethiopia Tigrinya, Spadine 2020; Eritrea Tigrinya, personal fieldwork) to Athabaskan (Slave, Rice 1986) and Turkic (Turkish, Şener and Şener 2011, Özyıldız 2012, Oguz et al. 2020; Uyghur, Sudo 2012, Shklovsky and Sudo 2014, Wang 2023; Chuvash, Knyazev 2022).

Just like logophoric pronouns, SIs are licensed in complex finite clauses headed by attitude verbs (Anand 2006; Oshima 2006; Sundaresan 2012, 2018; Deal 2020b); like logophors, their distribution does not seem to be subject to locality restrictions (Park 2014; Deal 2020b; Spadine 2020) and like them, they trigger agreement mismatches (Ganenkov and Bogomolova 2021; Ganenkov 2021; Knyazev 2022); last, they are preferably interpreted *de se*, just like logophoric pronouns (Schlenker, 2003; Anand, 2006; Sudo, 2012; Deal, 2020b).[9] In light of this, it is therefore plausible to assume (following the original proposal of Schlenker 2003) that such 'shiftable indexicals' are logophoric forms in the sense defined above, that is, first and/or second person elements specified with a -ACTUAL feature (77), and that languages displaying indexical shift make use of the hierarchy in (78).

(77) **First-person shiftable indexical**



(78) **Featural system of languages with 1P/2P shiftable indexicals**

a. 1st: [+AUTH, +PART, -ACT]

b. 2nd: [-AUTH, +PART, -ACT]

[9] Although variation exists, just like for logophors (see §2.2); some languages allow some of their indexicals to be read *de re* in attitude reports, such as the locative indexical *kine* 'here' in Nez Perce (Deal, 2019).

c. 3rd: [-AUTH, -PART, -ACT]

Crucially, logophoric languages differ from indexical shift systems in that in the former, the first person is the result of the morphological spellout of [+AUTH, +ACT] and [+AUTH, -ACT] on separate lexical items, whereas on the latter, [+AUTH, ±ACT] is fully syncretic.

As discussed in the previous section, variation in logophoric systems is derived by identifying on which person ±ACT lexicalizes; analogous variation can be found for indexical shifting languages. For instance, just as we can find LP-systems in which the second person is -ACT (Wan, Aghem) and systems in which it is +ACT (Ewe, Donno Sɔ), some indexical shifting systems can shift first but not second persons. Such is the case of the language Dene (Athabasakan; Northwest Territories, Canada),[10] in which 1st person is shiftable but 2nd is not, (79):

(79) Simon rásereyineht'u hadi
Simon 2SG.hit.1SG 3SG.say
'Simon$_i$ said that you$_{a(c^*)}$ hit him$_i$.'
[Dene, Rice 1986: (53)]

This provides evidence that this language actually makes use of a very similar feature system that the one observed for Ewe, in which 2nd person is specified with +ACTUAL, whereas 1st is -ACTUAL:

(80) **Featural system with shifty 1P and indexical 2P**

a. 1st: [+AUTH, +PART, -ACT]

b. 2nd: [-AUTH, +PART, +ACT]

c. 3rd: [-AUTH, -PART, -ACT]

Just like we find languages with only addressee logophors, but no speaker logophors, we find languages that have shifty second person, but unshifty first person; in these languages, second person pronouns can denote non-actual addressees, and first person are restricted to current speakers:

(81) li dad wɛl nɛnɛ **ɔny** ùsr ir el
3SG.F say.PST 3PL DEM **2SG** build.IMP 3SG.OBJ house
'She$_i$ said to them$_j$ you$_j$ build her$_i$ a house.'
(lit. 'She$_i$ said to them$_j$ you$_j$ build me$_i$ a house.)
[Adioukrou (Kwa; Ivory Coast), Hill 1995: (8)]

(82) ògwú úgâ ókêkitó ító íkíbé gwúñ kàñ ɔmɔ ìkâtùmú ìnyí
DEM mother be crying.PST cry say child 3SG.POSS 3SG tell.PST.NEG give
òwù yê íbé òwù kàgɔɔk ífít yì
2SG INTR say 2SG follow.NEG play play
'The mother$_i$ was crying and said: "My$_i$ child$_j$, did I$_i$ not tell you$_j$ not to join this dance group"?'
(lit. 'The mother$_i$ was crying and said her$_i$ child$_j$, did she$_i$ not tell you$_j$ not to join this dance group?')

[10] The language is also called *Slave*, but its speakers tend to prefer the former denomination. We thank Amy Rose Deal for mentioning this to us.

(83) ògwú énêrìèeñ òbê, òwù ‘nga kàñ ‘mgbɔ kèyí ìrè ‘mbùbàn,
DEM man say.PST 2SG mother 3SG.POSS time DEM be curse

tap nyî ɔmɔ
put.IMP give.IMP 3SG

‘The man$_i$ said "Mother$_j$, this time (even if) you$_j$ curse me$_i$..."’
(lit. ‘The man$_i$ said his$_i$ mother$_j$, this time (even if) you$_j$ curse him$_i$...’)
[Obolo (Niger-Congo; Cameroon and Nigeria), Aaron 1992: (22)-(23)]

The featural specification of the 2nd person in Adioukrou and Obolo is therefore similar to that found in the system of Pero, with the exception that these languages do not lexicalize the bundle corresponding to [-AUTH, +PART, -ACT] on a dedicated logophoric element distinct from 2nd person:

(84) **Featural system with indexical 1P and shifty 2P**

a. 1st: [+AUTH, +PART, +ACTUAL]

b. 2nd: [-AUTH, +PART, -ACTUAL]

c. 3rd: [-AUTH, -PART, -ACTUAL]

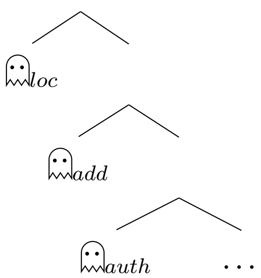
As a matter of fact, the Adioukrou/Obolo pattern is not predicted by current operator-based approaches to indexical shift (Anand and Nevins 2004; Anand 2006; Deal 2013, 2020b and much subsequent literature), in which shifting of indexicals is induced by the presence of a ‘monstrous’ operator 👻 in the embedded clause, which rewrites the Kaplanian context coordinates of a context-sensitive expression α with the values of i:

(85) a. $[\![👻\ \alpha]\!]^{g,c,i} = [\![\alpha]\!]^{g,i,i}$

b. $[\![$Kidane said that 👻 [I want to leave]$]\!]^{g,c,i}$ =
1 ⇔ $\forall i'$ compatible with what Kidane said in *i*, then the speaker in *i’* wants to leave in *i’*.

In order to capture variation, more sophisticated variations of the system have been proposed. Deal (2020b) proposes to expand the typology of 👻 so as they come into different varieties, depending on the kind of context parameter they can shift. She adopts a ‘cartographic’ approach where each operator appears in a dedicated position within the functional sequence:

(86) **Functional sequence of context-shifting operators** [Deal 2020b, (183)]



Lexical bundling is allowed between two adjacent operators within the hierarchy, but not between non-adjacent classes of 👻 within the sequence. For instance, the entire sequence can be bundled together to form the primitive 👻 that shifts all indexicals within its scope, (87a); similarly, $👻_{add}$ and $👻_{auth}$ can be bundled together to yield a $👻_{pers}$ that only shifts person indexicals, (87b). However, the system is designed so as to rule out any operator that would shift only the *addressee* while leaving the *author* coordinate untouched, as in (87d):

(87) **Varieties of context-shifting operators** [Deal 2020b, (177-178)]

a. $[\![👻\ \alpha]\!]^{g,c,i} = [\![\alpha]\!]^{g,i,i}$ (attested in Matses)

b. $[\![👻_{pers}\ \alpha]\!]^{g,c,i} = [\![\alpha]\!]^{g,<\mathbf{s(i)},\, \mathbf{a(i)},l(c),t(c)>,i}$ (attested in Uyghur)

c. $[\![👻_{auth}\ \alpha]\!]^{g,c,i} = [\![\alpha]\!]^{g,<\mathbf{s(i)},a(c),l(c),t(c)>,i}$ (attested in Slave)

d. $[\![👻_{add}\ \alpha]\!]^{g,c,i} = [\![\alpha]\!]^{g,<s(c),\mathbf{a(i)},l(c),t(c)>,i}$ (unattested?)

However, the Adioukrou and Obolo data outlined above would precisely require an operator of the sort defined in (87d), allowing shifting of 2SG while leaving 1SG unaffected; on the present approach, however, it is expected that ±ACT might be able to parametrize on different persons with no such restrictions, therefore carpturing the patterns in (81) and (82) for SI-systems, and those in (66)-(69) for logophoric systems. Additionally, the theory outlined here predicts global optionality in shifting, since every shiftable element bearing -ACT will always be able to obtain its reference via binding at the embedded level (shifted reading) or via the assignment function and the actual context c (indexical reading). We take this to be a welcome result, considering that indexical shift is by and large an optional phenomenon (cf. Sundaresan 2018).

In spite of this, however, such a unified theory of both phenomena faces non-trivial challenges, the biggest of which concerns so-called 'shift together' effects. The present theory predicts that every -ACT element should be able to shift independently, just like in the original proposal of Schlenker (2003); however, as noted by Anand (2006) and emphasized by Deal (2020b), such a system overgenerates, as it fails to account for the pattern illustrated in (88):

(88)	vɨzeri	Rojda	Bill-ra	va	kɛ	ɛz	to-ra	miradisa
	yesterday	Rojda	Bill-to	say.PST	COMP	1SG	2SG-to	angry.be.PRS

✓ 'Yesterday Rojda$_i$ said to Bill$_j$ that he$_i$ is angry at him$_j$.'
✓ 'Yesterday Rojda$_i$ said to Bill$_j$ that I am angry at you.'
✗ 'Yesterday Rojda$_i$ said to Bill$_j$ that I am angry at him$_j$.'
✗ 'Yesterday Rojda$_i$ said to Bill$_j$ that he$_i$ is angry at you.'
[Zazaki, Anand and Nevins 2004: (13)]

Zazaki being an optional-shifting language, the sentence in (76) is only two-way ambiguous: under a shifted interpretation, both first- and second-person indexicals refer to the reported speaker and addressee, respectively; under the unshifted interpretation, they refer to $s(c)$ and $a(c)$. Crucially, mixed or 'cross-contextual' readings are excluded. Any account (such as the present one) that does not place any binding restrictions on the index variable cannot straightforwardly derive such effects; by contrast, such a restriction follows from approaches that achieve shifting via a monstrous operator, since any indexical within the scope of 👻 will shift, regardless of its person specifications. This potentially suggests that indexical shift and logophoricity, though closely related, might in fact be two distinct phenomena – a route pursued by Deal 2020b, 2024).

4.5 Alternatives, competition and the internal structure of pronouns

So far, we hope to have motivated the following two claims: i) pronouns denote complex structures (Cardinaletti, 1994; Cardinaletti and Starke, 1999; Déchaine and Wiltschko, 2002; Patel-Grosz and Grosz, 2017) and ii) they display markedness asymmetries that can be accounted for by the meaning of their features (Noyer, 1992; Harley and Ritter, 2002; Sauerland, 2008). As argued in §4.1, such meanings are presuppositional and can be interpreted as partial functions restricting the pronominal index of a given pronoun. Elements of different presuppositional strenght are expected to compete with each other (Hawkins, 1991; Heim, 1991), with a preference for elements with a stronger presupposition. Heim's *Maximize Presupposition!* predicts just that: elements of greater presuppositional strength should block use of weaker elements of equal assertive strenght, (89):

(89) **Maximize Presupposition! (standard version; to be revised)**
Do not use ϕ in context C[11] if there is a $\psi \in \textsc{alt}(\phi)$ s.t.

a. the presuppositions of ψ and ϕ are satisfied within C;

b. $[\![\psi]\!]^C = [\![\phi]\!]^C$, and

c. the presupposition of ψ asymmetrically entails the presupposition of ϕ.

Taken as a pragmatic filtering condition on utterances, the principle states that, given a presuppositional element ϕ that has a set of alternatives $\textsc{alt}(\phi)$, speakers should prefer to use any member of that set ψ if it is (i) presuppositionally stronger, and (ii) true in the context of utterance. If a competent and cooperative speaker were to utter ϕ under those conditions, then the hearer would consistently infer that she did not utter the presuppositionally stronger ψ on purpose, and that the speaker does not know whether ψ is the case or not: in other words, the utterance of ϕ would give raise to an *antipresupposition* (Percus, 2006), as in (90):[12]

(90) *Context: John is speaking to Mary.*

a. $^{\#}$John is happy.

b. I am happy.

c. $^{\#}$Mary is happy.

d. You are happy. (adapted from Schlenker 2005: (18))

While in that context, both *John* and *I* refer to the speaker, and *Mary* and *you* to the addressee, sentences involving proper names instead of indexicals are perceived as deviant. As previously argued by Schlenker (2005) and Marty (2017), if uttered in a context where John is the speaker and Mary is the addressee, sentences (90a) and (90c) will be perceived as odd because in that context, indexicals *I* and *you* are favored by *MP!* over proper name DPs if they are meant to refer to the same individual. Under this approach, disjointness

[11] In what follows, following standard usage, we use capital C here to denote the stalnakerian context set (Stalnaker, 1974), that is, the set of all possible worlds compatible with the *common ground*, that is, the set of all possible propositions compatible with what the interlocutors in a conversation believe/take for granted and not subject for further discussion. This is to be contrasted with the Kaplanian context c used so far.

[12] It is commonly accepted that this inference is eventually strengthened somehow, leading the hearer to infer that the speaker does not believe ψ to be true (Spector, 2003; Sauerland, 2004b,a; Chemla, 2008).

inferences are the result of feature competition: if two pronominals compete for reference with another DP, then use of the presuppositionally weaker element leads to its indexing to another compatible referent:

(91) *Context: same.*

a. He is happy.
$\rightsquigarrow$ The referent of *he* is not the actual speaker or addressee.
$\rightsquigarrow$ The referent of *he* and John are distinct individuals.

b. $\text{ALT(91a)} = \left\{ \begin{array}{c} \text{I am happy} \\ \text{you are happy} \end{array} \right\}$

The disjointness effects mentioned in §2.4, we argue, are cases analogous to (91). Laid out in informal, Gricean terms, the idea is quite simple: when reporting what someone said, a speaker *s* of a logophoric language *L* is expected to use a first-person form whenever the reported speaker (the subject of the matrix clause) is intended to co-refer with the subject of the embedded clause. If the speaker uses a 3rd person form instead, then they antipresuppose that both forms do not co-refer, so their referents must be distinct individuals centers (i.e. distinct context-individual pairs). In order to derive such inferences, the competition algorithm in (89) needs to be refined, however, because we ultimately want the presuppositions of pronouns to be computed not only against the set of the *actual* context and common ground, but the set of possible contexts that the attitude verb quantifies over. Here we follow a suggestion by Stalnaker (2014) to understand the common ground not merely as a set of possible worlds, but as a set of K(aplanian)-contexts - that is, centered worlds containing time and place parameters as well. The relevant competition mechanism for antipresuppositions should be adjusted in order to refer to this augmented notion of common ground, the set of all possible K-contexts κ:

(92) ***Maximize presupposition!* (relativized to possible indices)**
Do not use ϕ with respect to the current common ground C and assignment g if $\exists\psi \in \text{ALT}(\phi)$ such that

a. $\forall i \in C, \phi \in dom([\![\cdot]\!]^{g,c,C})$ and $\psi \in dom([\![\cdot]\!]^{g,c,C})$

b. $\forall i \in C, [\![\phi]\!]^{g,c,C} = [\![\psi]\!]^{g,c,C}$, and

c. $\forall i \in \kappa$, if $\psi \in dom([\![\cdot]\!]^{g,c,C})$, then $\phi \in dom([\![\cdot]\!]^{g,c,i,C})$, but not the other way around.

This revised statement of *MP!* allows us to enforce competition among alternative utterances with different presuppositional strenghts across possible indices κ, thought of the set of possible contexts: for two alternatives ϕ and ψ, using ϕ over ψ will be infelicitous if i) both have their respective presuppositions satisfied in C, ii) both are denotationally equivalent in C, and iii) the presuppositions of ψ asymettrically entails the presuppositions of ϕ across every possible index $i \in \kappa$.

4.5.1 Accounting for disjointness inferences

We are now ready to derive the disjointness inferences mentioned in §2.4. First, recall that, in logophoric contexts, a 3rd person pronoun cannot be used *in lieu* of a LP if the intended referent is the logophoric center:

(14) a. Nnsɨni dzɛ enyɨa é bvʉ nù
Nsen say COMP LOG fall FOC
'Nsen$_i$ said that she$_i$ fell.'

b. Nnsɨni dzɛ enyɨa ù bvʉ nù
Nsen say COMP 3SG fall FOC
'Nsen$_i$ said that she$_{*i/j}$ fell.'

[Aghem, Butler 2009: (10-11)]

Given (92), a disjointness inference is indeed predicted for (14b), since (14a) is an alternative to it and has a stronger presupposition, due to +AUTH. Since the sentence in (93a) has a presuppositionally stronger alternative compatible with the context, (93b), then *MP!* dictates that it should be used, on pains on triggering oddness.[13]

(93) a. #Nsen$_i$ said that 3SG$_i$ fell.

b. ALT(93a) = $\{$ Nsen$_i$ said that LOG$_i$ fell $\}$

In a similar fashion, the above predicts the *1-LOG pattern: in cases in which the speaker wants to refer to herself in a logophoric context (i.e., when the matrix subject is first person), then, by *MP!*, a first person pronoun has to be used in the embedded clause for co-reference, since it is presuppositionally stronger: +ACT restricts its referent to the center of c, the actual context.

(18) a. Kofí ŋa bə yi lɔ Áma
Kofi know COMP LOG love Ama
'Kofi$_i$ knows that he$_{i/*j}$ loves Ama.'

b. #ŋ̀ə ŋa bə yi lɔ Áma
1SG know COMP LOG love Ama
Intended: 'I$_i$ know that I$_i$ love Ama.' [Danyi Ewe, O'Neill 2015: (3a, c)]

(94) a. #I$_i$ know that LOG$_i$ love Ama.

b. ALT(94a) = $\{$ I$_i$ know that 1SG$_i$ love Ama $\}$

The same mechanism also accounts for the fact that regardless of their person specifications, 3rd as well as 2nd person pronouns can serve as antecedents of logophors: since they are both non-authors in c but authors in the embedded context i, the pronoun with the stronger person presupposition should be used for cross-reference – in that case, the LP:

(19) a. là gé ɓà súglù é lɔ
2SG said LOG.SG Manioc DEF ate
'You$_i$ said you$_i$ had eaten the manioc.'

b. à gé mɔ kú má
2PL said LOG.PL house EQUAT
'You$_i$ said it was your$_i$ house.' [Wan, Nikitina 2012a: (5a, b)]

[13]Note that the sentence in (93a) has other structural alternatives (cf. *infra*): *Nsen said that I fell, Nsen said that you fell*. However, these alternatives have different meanings (due to different indexing) and therefore, do not compete with (93a).

(95) a. #You$_i$ said 2SG$_i$ had eaten the manioc.

b. ALT(95a) = { You$_i$ said LOG$_i$ had eaten the manioc. }

The present account therefore explains the *raison d'être* of standard cases of 'person neutralization' mentioned in the typological literature (Von Roncador, 1992; Curnow, 2002b,a).[14]

4.5.2 Alternatives and features

A crucial component of the *MP!*-based approach to antipresuppositions (as well as other implicature-related phenomena) is the definition of the alternative set ALT, over which the inference mechanism operates. Most of the proposals in the literature follow neo-griceans accounts such as that of Horn (1972) and Gazdar (1979) in positing scales, in which elements of a given scale are ordered with respect to one another in a monotonic fashion. On these accounts, *MP!*-based inferences can be predicted to arise because the presupposition triggers they involve are scalemates, and any utterance of the weaker element of the scale is likely to generate an antipresupposition about the stronger element. However, as argued forcefully by Katzir (2007) and Fox and Katzir (2011), scalar approaches to alternatives give raise to considerable problems, the most prominent of which being their very nature and origin. In order to avoid these, we follow Rouillard and Schwarz (2017) in adopting Katzir (2007)'s theory of structural alternatives for the presuppositional domain. In Katzir's account, structural complexity plays a crucial role in determining what counts as an alternative for a given sentence ϕ; the set of formal alternatives ALT(ϕ), represented in (97) will consist in that set of alternatives that are at-most-as-complex as ϕ, which are created by replacing constituents of ϕ with elements in the subsitution source of the language, (99):

(97) $\text{ALT}(\phi) = \{\psi : \psi \precsim \phi\}$

(98) **Structural complexity** [(Katzir, 2007, (19))]
Let S, S' be parse trees. S' can be said to be at-most-as-complex as S if we can transform S into S' by

[14] A reviewer brought to my attention the case of language Jula (Niger-Congo, Mande: Burkina Faso, Ivory Coast and Mali), which does not seem to display the pattern exemplified in (19):

(96) #i ye a fɔ ko ale facɛ na-na
2SG PFV 3SG say COMP 3EMP father come-PFV
Intended: 'You$_i$ said that your$_i$ father has come.' [Jula, Kiemtoré 2022: 213, (28b)]

Here, the emphatic third person form *ale*, which can be used as a logophor in Jula complex clauses headed by *ko*, cannot take a second person pronoun as antecedent, thus escaping the generalization in (95). We would like to suggest that this is possibly due to the fact that *ale* is not a genuine logophor, but (as emphasized by the author himself in his gloss) a third person pronoun with an emphatic meaning that is merely 'recruited' in logophoric contexts. As a consequence, it might be the case that *ale* bears no +AUTHOR feature at all, therefore not entering competition as in example (19). This is actually corroborated by Kiemtoré (2022), who comments this example as follows: "Recall that there is no restriction on the person features of the source DP within *ko*-clause complementation. Therefore, the infelicity of the sentences in (96) is due to a clash in person-feature between *ale* and the source DP. The former being a third-person pronoun, it cannot take first and second-person DP as antecedents." We should therefore not expect languages with 'indirect logophors' such as Jula to obey the pattern discussed above.

a. deleting constituents of S,

b. contracting (i.e., merging and identifying nodes) constituents of S,

c. replacing constituents of S with constituents of the same category from the Substitution Source of the language.

(99) **Substitution Source (SS)** [(Katzir, 2007, (41))]
Let ϕ be a parse tree. The substitution source for ϕ, written as L(ϕ) is the union of the lexicon of the language with the set of all substrees of ϕ.

This felicitously derives the fact that structurally more complex alternatives of a sentence *S* are generally not part in the alternative domain ALT(ϕ) and therefore, cannot be negated during implicature computation.

Since the substitution source (SS) in (99) that feeds the alternative-generating mechanism is defined with respect to the lexicon, a question immediately arises when it comes to pronominal reference and the related data presented in this paper. If one takes at face value person markedness asymmetries as well as the structural view on pronouns adopted in §4.1, then one might wonder whether the algorithm in (98) should be defined with respect to the lexicon exclusively (as the set of atomic vocabulary items a language disposes of), or rather, with respect to parse trees only, albeit below the world level. The first option would require to amend the complexity-based algorithm in (98) so as it defines the lexicon not with terminal nodes within the syntax, but as syntactic words. There are reasons, however, not to pursue this route, since an appropriate notion of 'syntactic word' is still to be defined (see discussions in Haspelmath 2011 and Svenonius 2018, 2020). Another, more promising option would be to relativize (98) to parse trees below the word level and assume that morphosyntactic complexity at the level of features is relevant when computing strenghtening inferences of the kind that *MP!* leads to (Sauerland and Bobaljik, 2022; Alexiadou et al., 2024; Haslinger, 2026). Consequently, looking into the feature matrices of pronouns, we can see that our system ensures that the pronominal forms of a language with speaker logpophors such as (75) are of equal complexity, allowing them to be alternatives to each other and therefore compete by the algorithm outlined in (92).

(75) **Featural system with speaker LOG and +ACTUAL 2P (Ewe, Donno Sɔ)**

a. 1st: [+AUTH, +PART, +ACT]

b. LOG: [+AUTH, +PART, -ACT]

c. 2nd: [-AUTH, +PART, -ACT]

d. 3rd: [-AUTH, -PART, -ACT]

An analogous representation of the same paradigm with privative features would eschew the complexity algorithm and, as a consequence, would predict no disjointness inference to arise, contrary to fact:

(100) a. 1st: [AUTH, PART, ACTUAL]

b. LOG: [AUTH, PART]

c. 2nd: [PART, ACTUAL]

d. 3rd: $\emptyset$

In such a paradigm, pronouns get increasingly more complex as they are specified with more features, preventing LOG to count as an alternative to the 3rd person, and the 1st person to count as an alternative to LOG. Note that the problem generalizes to all current privative accounts of pronouns, and arises independently of whether or not we posit ±ACTUAL.

Therefore, given structural complexity and taking the various inferences observed in §2.4 as a test for probing the morphosyntactic structure of the elements that compete in a given language, the above considerations represent an argument in favor of the bivalence of person features, as opposed to privativity (Harbour, 2011a, 2013).[15] Relatedly, adopting a morphosyntactic complexity-based approach to pronominal anaphora also argues against a definition of LPs as third person pronouns *cum* a logophoric feature, a view adopted in almost all previous accounts, including Heim (2001, 2002); von Stechow (2002, 2003); Baker (2008); Pearson (2015) and Bassi et al. (2025): on this view as well, LPs are more complex than 3rd person pronouns, and therefore eschew competition on similar grounds. Additionally, all these accounts inherit the burden of accounting for the typological reality of the LOG feature, and the questions it presses: is LOG a person feature? If yes, how does it interact with other features across paradigms? If not, what kind of feature is it, and why is it to be found only in the pronominal domain? By contrast, the present account, in which ACT suffices to derive most, if not all, logophoric systems, is fully conservative with respect to person inventories posited so far for natural languages.

Logophoric paradigms	**Present system**	**Bassi et al. (2025) system**
1st	[+AUTH, +PART, +ACTUAL]	1
LOG	[+AUTH, +PART, -ACTUAL]	[3, LOG]
2nd	[-AUTH, +PART, -ACTUAL]	2
3rd	[-AUTH, -PART, -ACTUAL]	3

Table 6: A comparison of the featural hierarchies of the present system vs. Bassi et al. (2025).

5 Conclusion

In this article, we have tried to provide a detailed account of the morphosemantic properties of a given class of pronouns, the logophoric pronouns found in West-African languages, the properties of which we argued are best explained by considering them full-fledged first-personal elements, albeit non-indexical ones. We have done so by proposing that pronouns in general are complex elements denoting individual concepts, the denotations of which are presuppositionally restricted by two kind of features: person features (±AUTH and ±PART), as well as an additional ±ACT which specifies whether a pronominal element will be able to be bound or left free by attitude verbs in a given language. We have then showed that the existing range of logophoric systems is correctly accounted for

[15]Note that an alternative, privative solution would be to allow the syntax to represent a semantically vacuous - but syntactically contentful - PERSON feature on the 3rd person; this is essentially the strategy pursued by Alexiadou et al. (2024), on the basis of the (un)availability of generic readings in presence/absence of definite determiners in Romance and Germanic languages. The choice of one or the other solution therefore depends on one's own take about arguments for or against feature privativy; for a review of these arguments, see notably Harbour (2013).

by positing variation in the quantificational force of attitude verbs (which specifies, for a given language, the 'logophoric context' in which a LP can appear or not) as well as variation in how ±ACT lexicalizes alongside different person specifications to yield the attested logophoric forms across languages. The present account combines various insights of its predeccessors, borrowing from Schlenker (2003) and Deal (2024) the first-personal status of LPs and from Bassi et al. (2025) the fact that they are presuppositionally-restricted individual concepts, but ultimately departs from them in proposing that LPs consist of person-feature bundles similar to other pronouns, competing with other forms in a given paradigm to yield their attested distribution. In addition, we argued that the various inferences that these pronouns give raise to are best explained by looking at their internal morphological complexity, vindicating an approach to alternatives defined not in terms of lexical elements (as in Katzir 2007) but in terms of syntactic terminals only, with important consequences for the theory of person features (Harbour, 2011b,a, 2013).

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